



COMPUTER AIDED ANALYSIS AND DESIGN

Course (DE ZG622) - Lecture 1

Prof. Ruchi More Kotecha

Course Outline



- Introduction to Product lifecycle, CAD/CAM systems and Geometric Modeling
- Introduction to modeling in Creo
- 2D and 3D Geometric transformations
- Geometric Modeling of Analytic and Synthetic Curves
- Bezier and B-Spline Curves
- Machining Application of Creo
- Modeling of Analytic and Synthetic Surfaces
- Solid Modeling
- Assembly Modeling
- Data Exchange
- Computer Aided Analysis

Evaluation Scheme

Evaluation Scheme

Evaluation Component	Name	Type	Weight	Duration	Schedule
<i>EC - 1</i>	<i>Assignments</i>	<i>Individual and Take-home</i>	<i>10%</i>	<i>1 Week</i>	
EC-1	Lab	Online	20%	2 Weeks	
<i>EC - 2</i>	<i>Mid-Semester Examination</i>	<i>Closed Book</i>	<i>30%</i>	<i>2 Hrs</i>	
<i>EC - 3</i>	<i>End-Semester Examination</i>	<i>Open Book</i>	<i>40%</i>	<i>2.5Hrs</i>	

Syllabus for Mid-Semester Test (Close Book): Topics in Session Nos. 1 to 8

Syllabus for Comprehensive Exam (Open Book): All topics (Session Nos. 1 to 16)

Note: This information is subject to change, and the changes, if any, will separately be informed to the students.

Key Takeaways



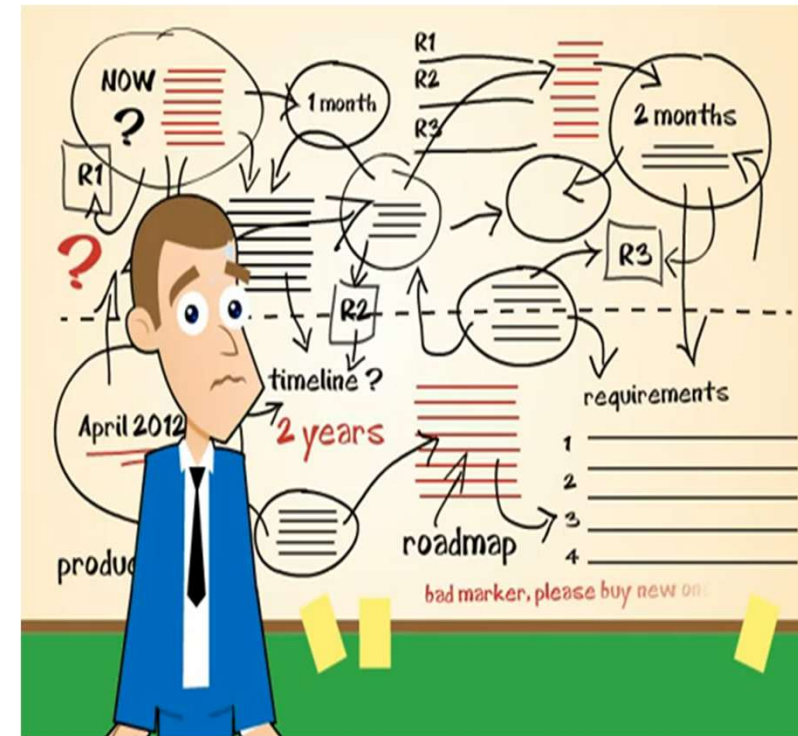
- Understand the role of CAD/CAM/CAE in Product Design and Overall Product Lifecycle
- Concepts of Geometric Modeling
- Mathematical Concepts Behind Modeling Geometric Entities like Curves, Surfaces and Solids
- Modeling and Representation of Solid
- Modeling of Assembly
- Exchange of Data Within CAD Systems
- Introduction to FEA





Today's Industry's Challenges

- **Shorter** product life cycle (from years to months)
- **Increased** product complexity (interdisciplinary systems embedded)
- **Changing** consumer preferences driving **frequent** design and model changes
- **Greater** variety and **smaller** volumes
- **Increased** demand for productivity and product quality
- **Lesser** time to market
- Financial **fluctuations**
- **Pressures** of cost, margin and market share



Disruption in Mobility



1900 - Where is the car?

1913 - Where is the horse?

2030 ??





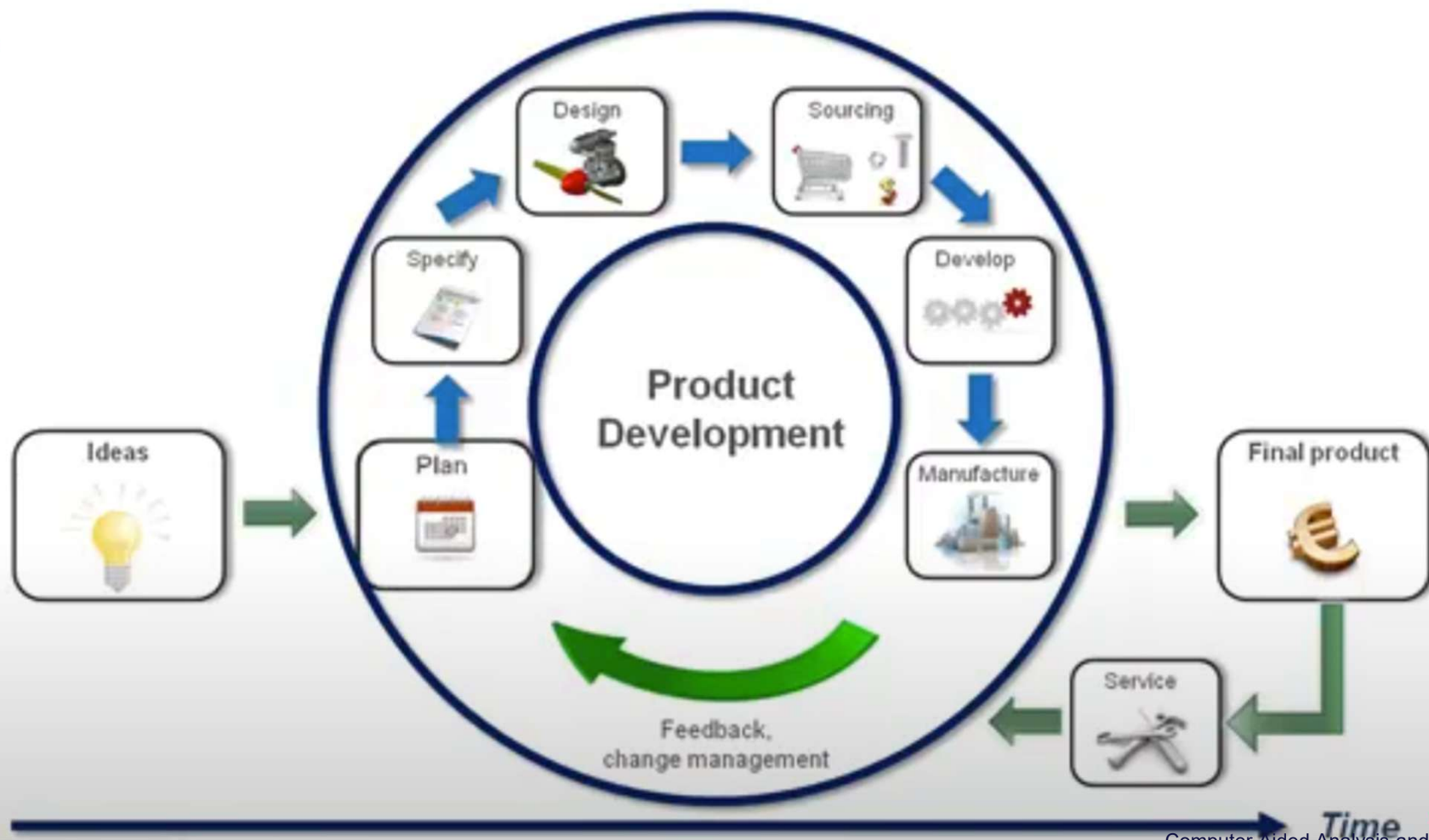
Significance of Issues Related to Product

- “Core” of the business
- Out of 100 product development projects, 63 are cancelled, 25 commercially succeed and 12 fail (Booz Allen)
- Up to 1/3 of product development time is spent on unnecessary work or waiting for decisions (McKinsey)
- 46% of product development resources are spent on products those fail (Cooper Product Development Institute)
- The revenue and direct costs directly proportional to product competitiveness which is a function of design
- Up to 25% time is spent on searching for correct product data
- Products often undergo change and tracking changes is very critical

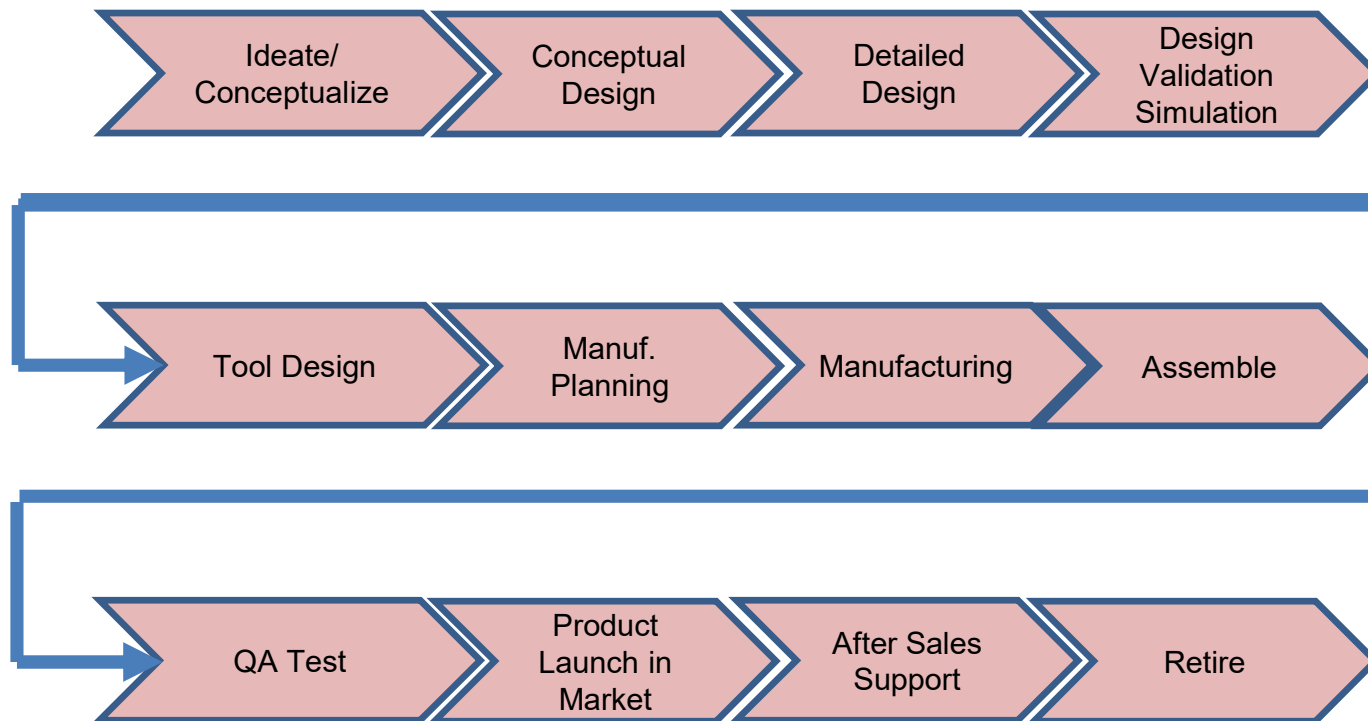


Typical Product Lifecycle

Innovation to a Final Product



Product Development & Lifecycle



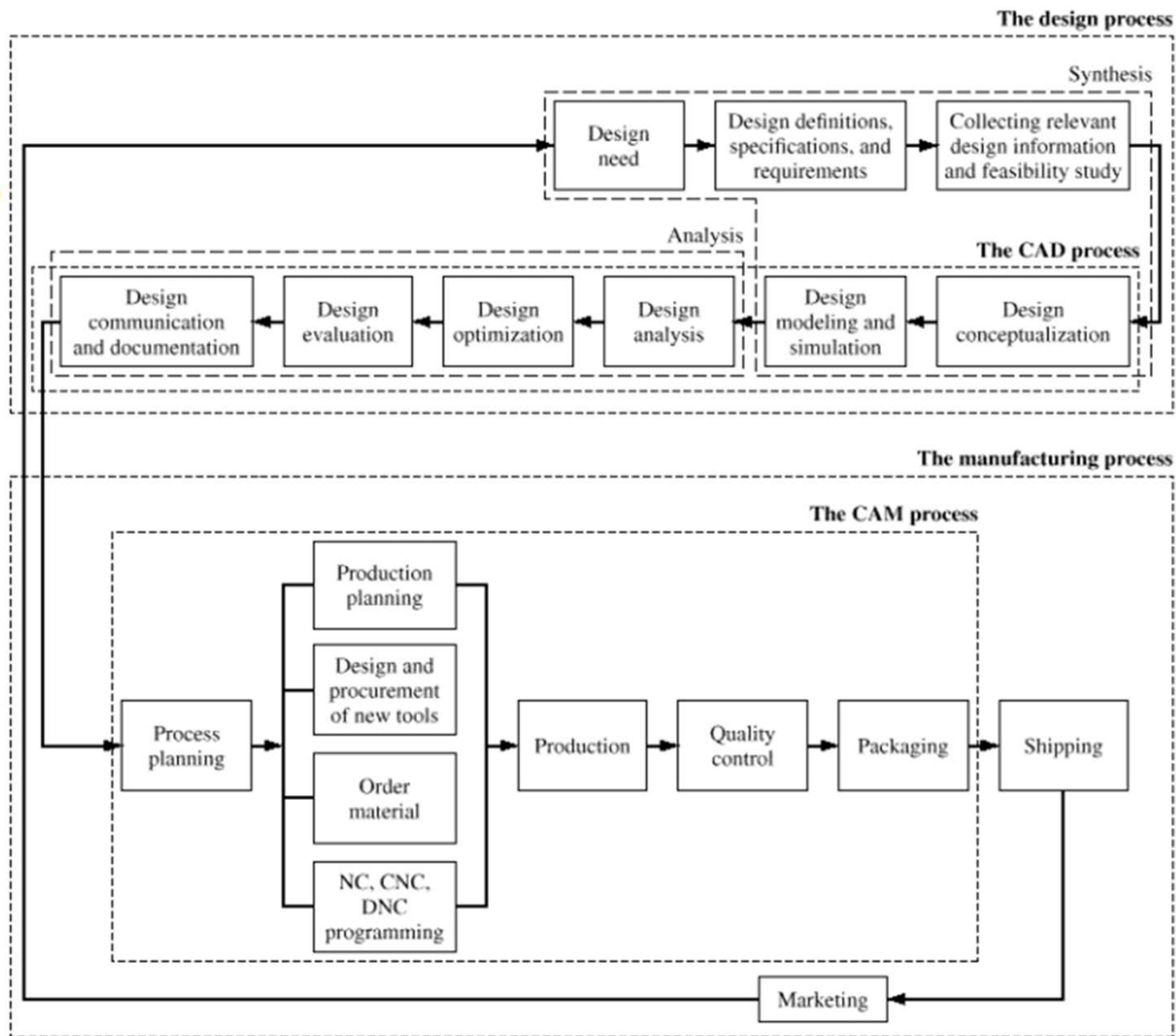


Figure 1.1 A typical product cycle.

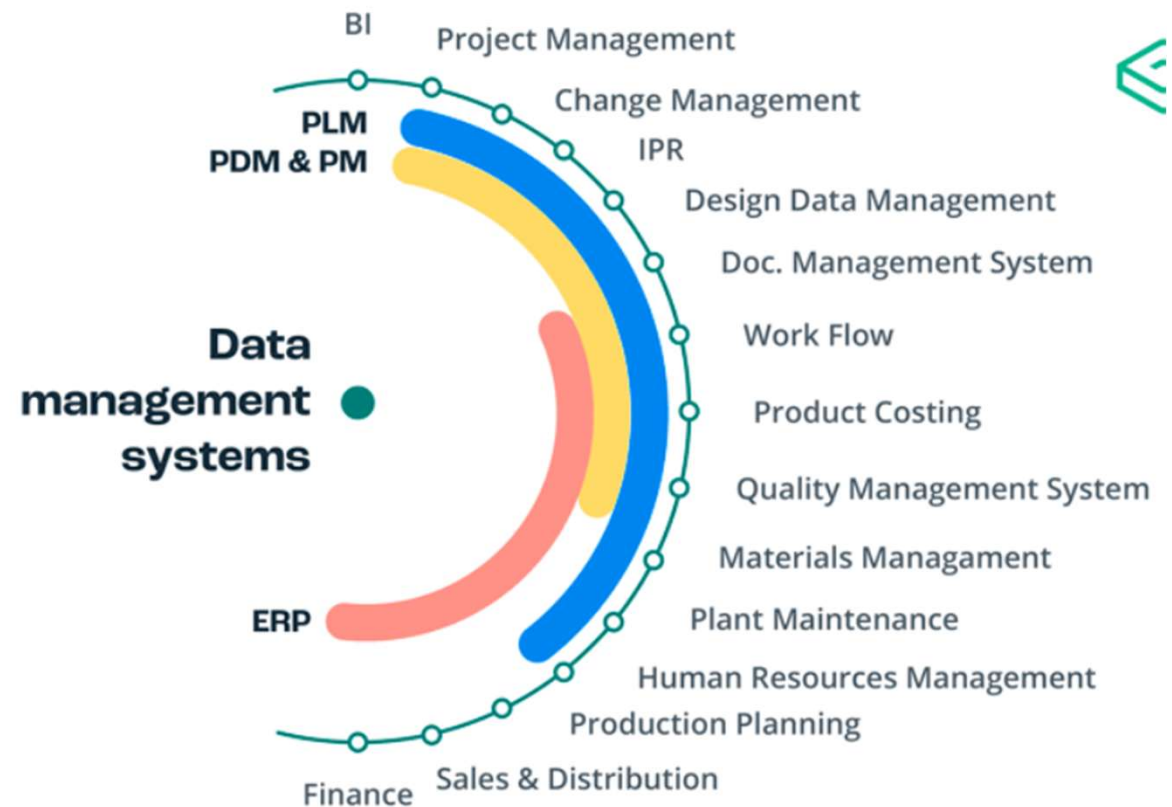
PDM vs ERP vs PLM



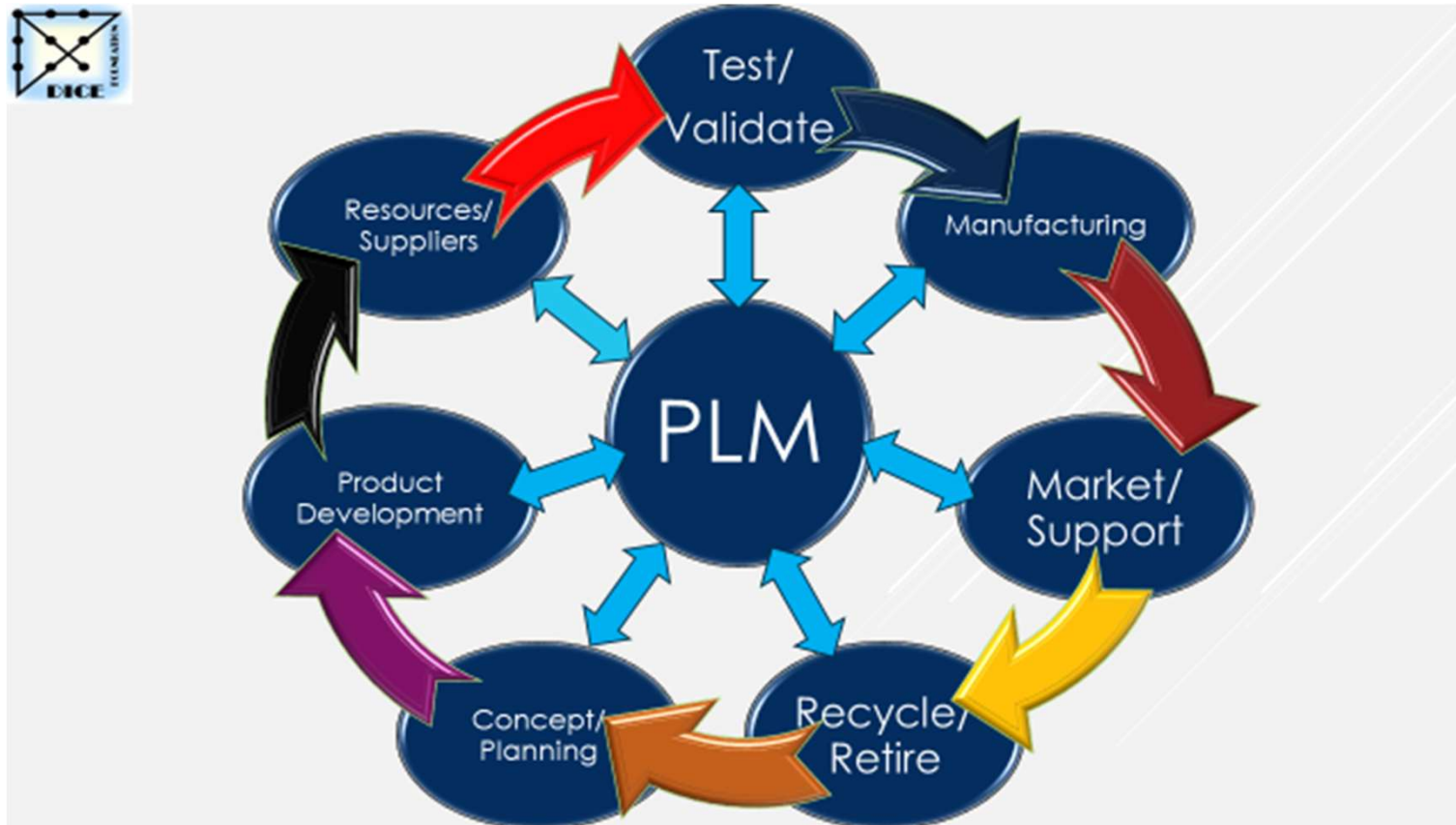
What is PDM?

What is ERP?

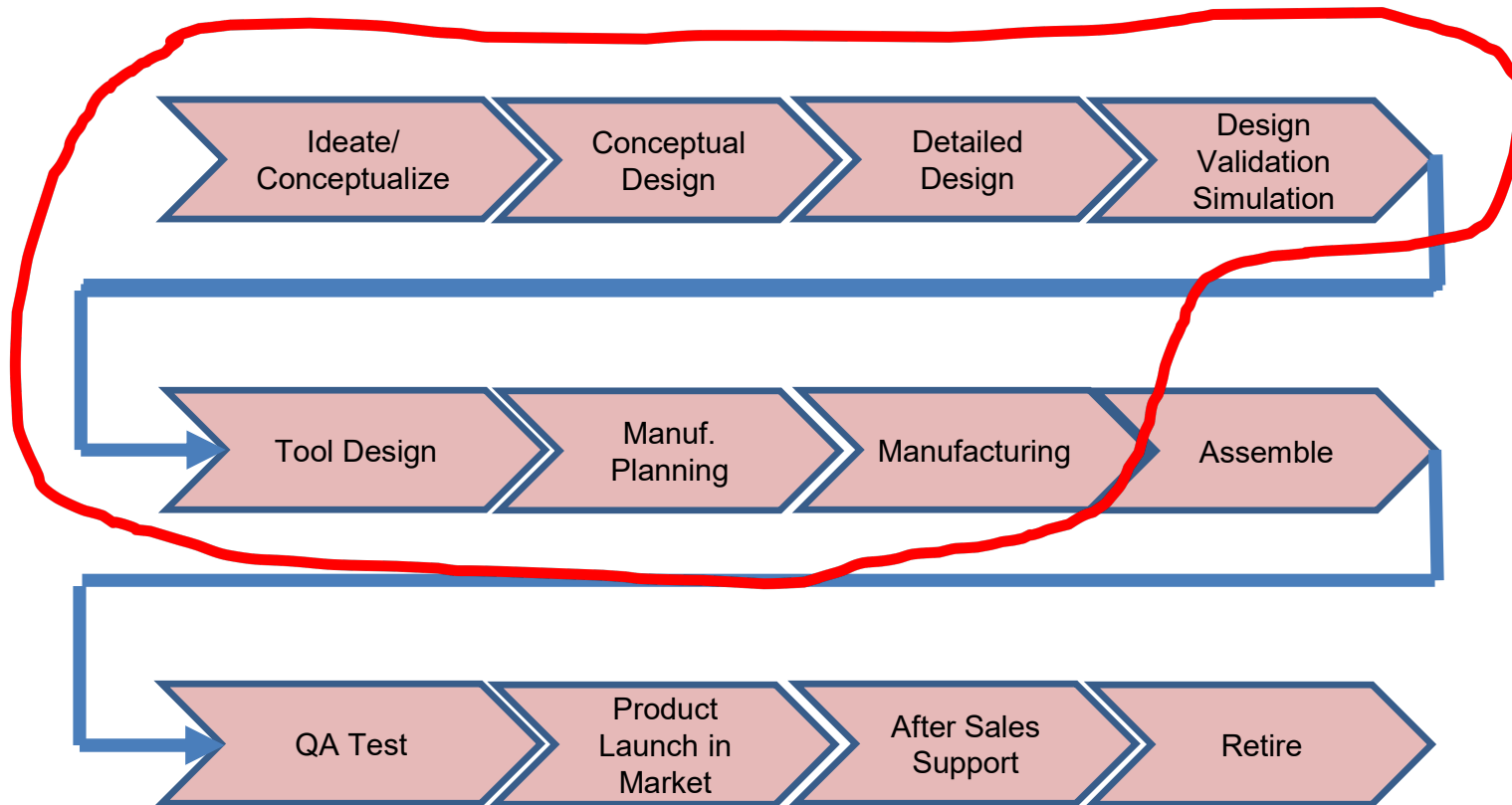
What is PLM?



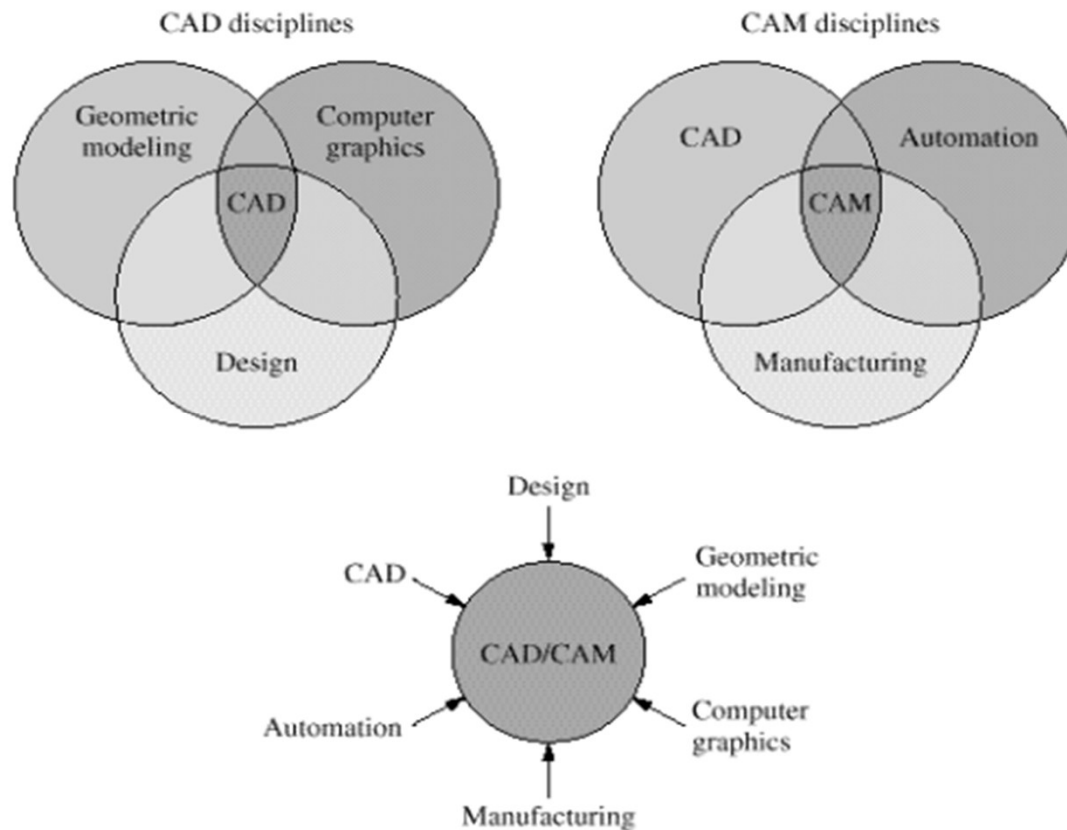
Product Lifecycle Management (PLM)



CAD/CAM in Product Lifecycle



CAD CAM in Product Lifecycle





CAD CAM and CAE

Computer Aided Design (CAD)



- Express product design ideas using 2D/3D sketches/models
- Model a component with complex shapes
- Represent solid geometry
- Enable modeling of complex curves and surfaces
- Model an assembly of components
- Render and animate the appearance and functionality of a product



Computer Aided Manufacturing (CAM)

- Use of computers to perform manufacturing operations.
- Direct or indirect interfacing of computers with Manufacturing Machine Tools through Numerical Control
- Programming of robots

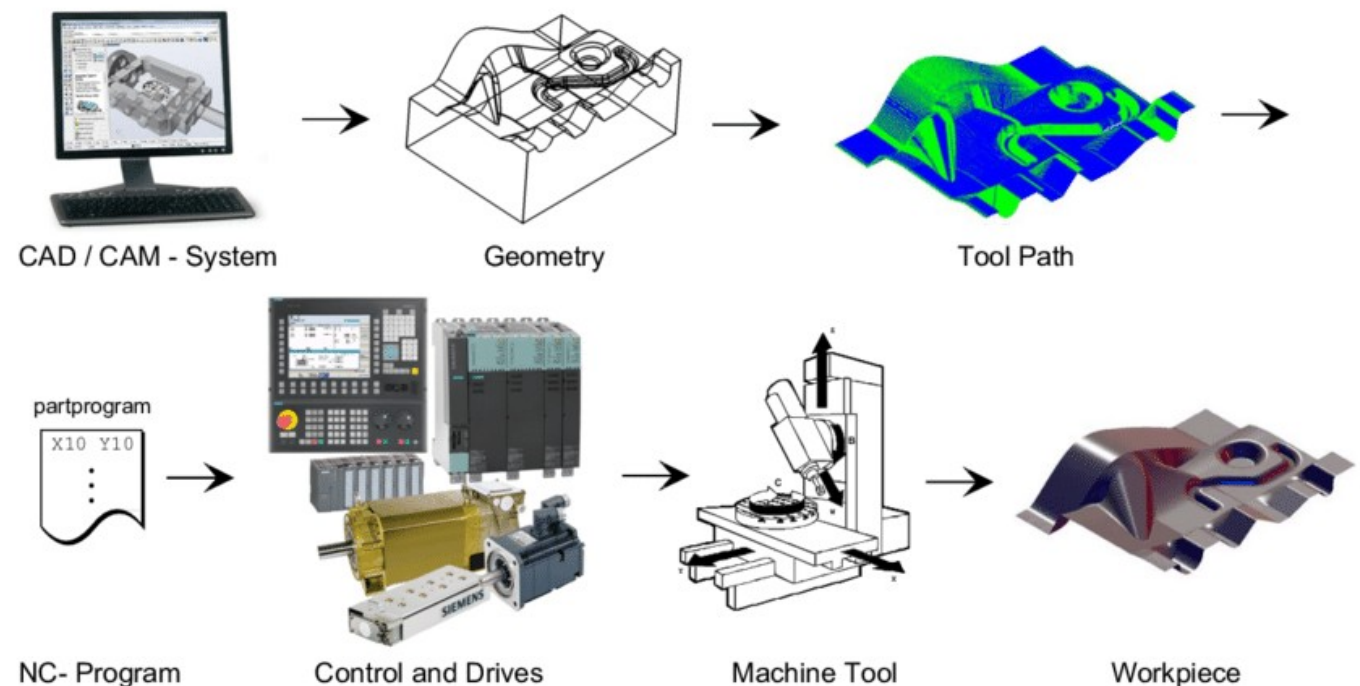
Computer Aided Engineering (CAE)



- Use of computers to analyse geometry.
- Allows designer to simulate and study the behaviour to help in optimisation
- Finite Element Method (FEM)
- Perform structural, Thermal analysis
- Perform kinematic motion study analysis
- Perform computational fluid dynamics analysis
- Design optimization using simulation

Overview of CAD CAM Systems

- Modeling/
Manufacturing
Features with UI
- Kernel /
Computation
Algorithms
(Geometric Engine)
- Storage (Local
Disk/Database)



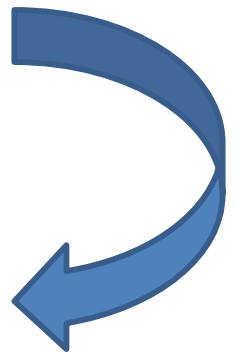
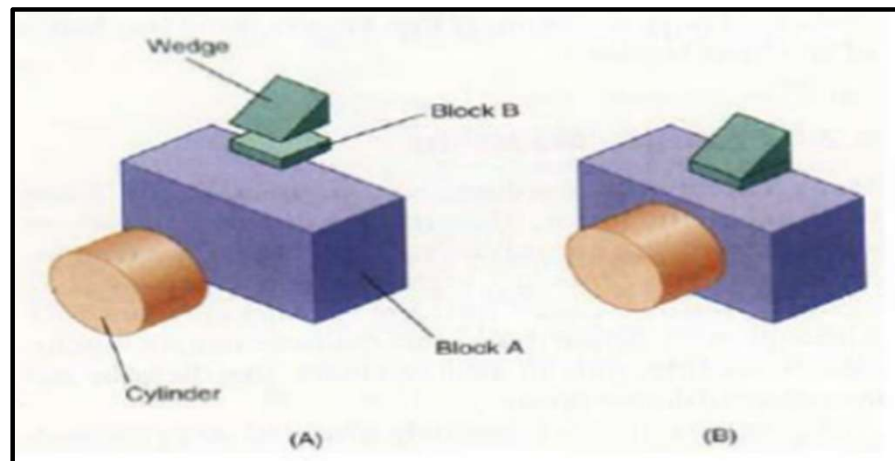
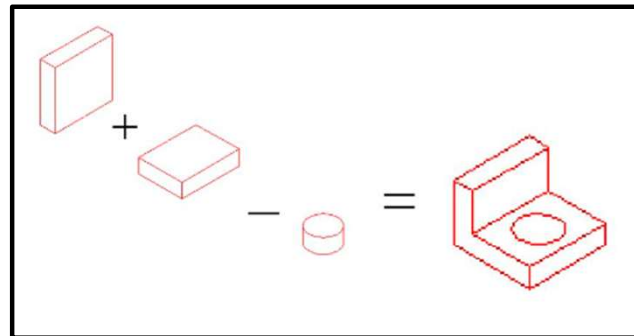
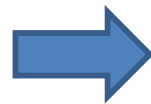
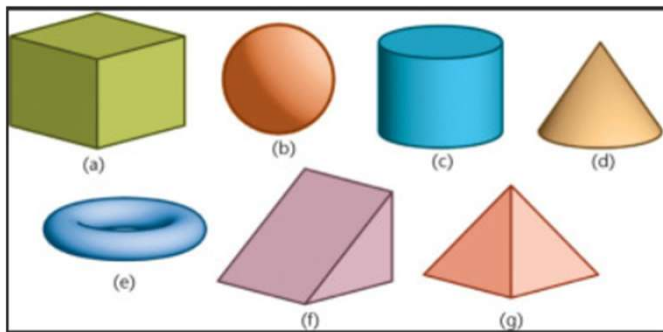
Geometric Modeling



- Creating/manipulating and digitally representing geometric entities (1D, 2D, 3D)
- Representation of Solids
- Inside/Outside, Boundaries
- Water-tight
- Topological and Geometric elements

-
- The image displays a mechanical component in two forms: a 3D perspective view on the left and a 2D technical drawing on the right.
- 3D Model (Left):** A yellow-colored mechanical part. It features a rectangular base plate with four circular mounting holes. A vertical support structure rises from the center of the base, consisting of a triangular web and a cylindrical top section with a large central opening.
- Technical Drawing (Right):** A detailed 2D orthographic projection of the part, showing top, front, and side views with dimensions in millimeters.
- Top View:** Shows the base plate with overall dimensions of 120 mm (width) by 90 mm (depth). The base has a thickness of 15 mm. There are four holes, each with a diameter of $\phi 10$. The distance between the centers of the front two holes is 35 mm. The distance from the front edge to the center of the front holes is 50 mm. The base has a central rectangular cutout with a width of 60 mm and a depth of 30 mm.
 - Front View:** Shows the vertical support structure. The base of the support is 30 mm wide. The web of the support has a thickness of 10 mm. The total height of the support structure is 89 mm. The top cylindrical section has an outer diameter of 100 mm and an inner diameter of 60 mm. The web is centered within the cylindrical section.
 - Side View:** Shows the profile of the support structure, confirming the 10 mm web thickness and the 89 mm total height.

Modeling with Solid Primitives

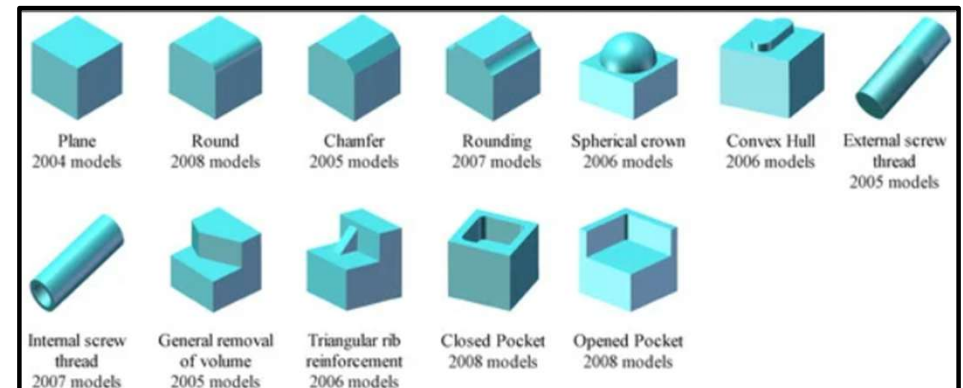
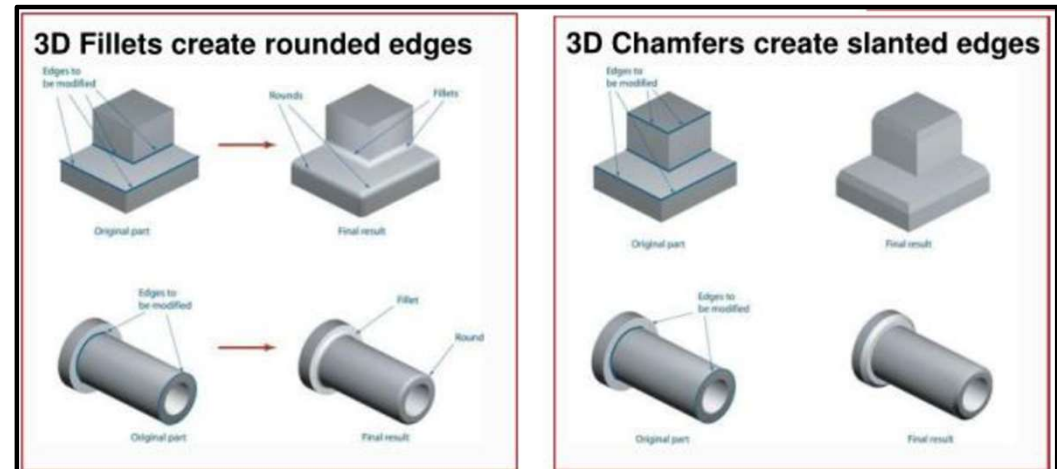


Feature Based Modeling

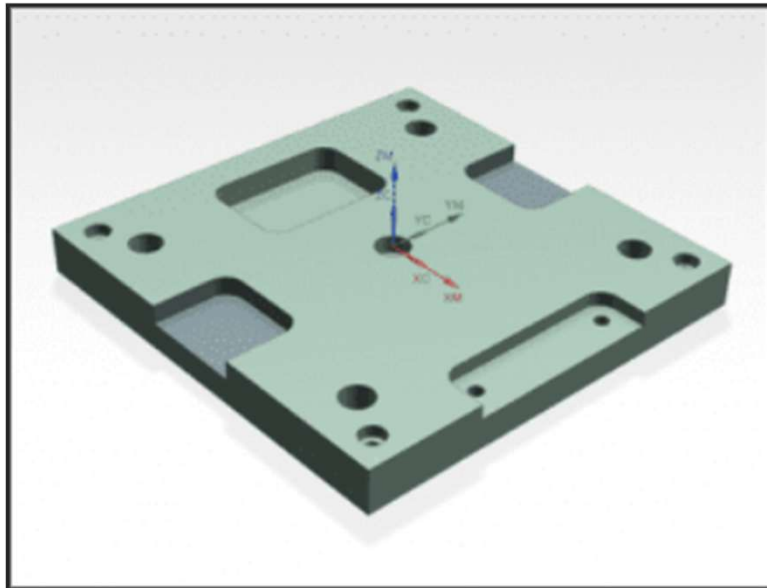


Using predefined geometric features

- **Edge based**
 - Chamfer
 - Fillet (Blend)
 - Step
- **Face based**
 - Hole
 - Pocket
 - Slot
 - Boss
 - Pad
 - Draft
- **Others**
 - Rib
 - Thread
 - Spring

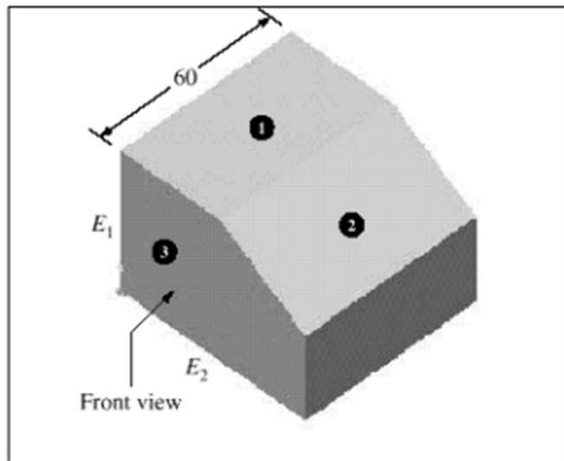


Feature Based Modeling

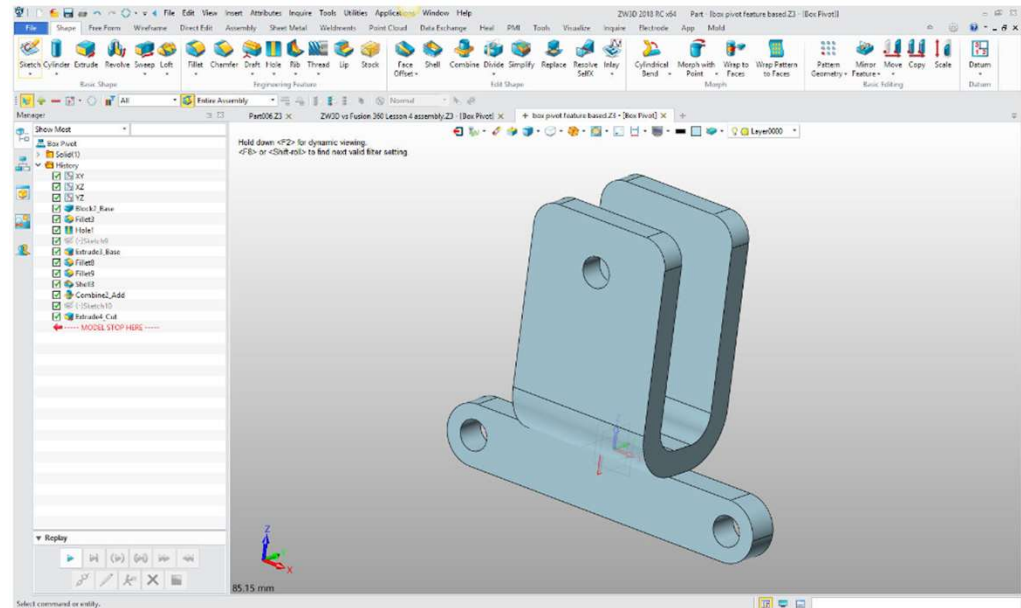
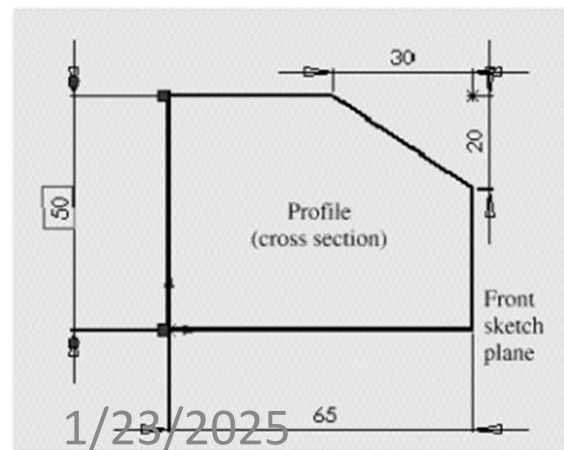


Isolated Generic Feature	Vertex-Class						
	Edge-Class	Edge Partition					
		Edge Elimination					
	Face-Class	Face Partition					
		Internal Loop					

Sketch Based Modeling



- Create 2D sketch on datum plane
- Specify dimensions
- Extrude/revolve/Sweep
- Boolean



Geometric Model Types

- 2-½ D
- 3D
- Combined

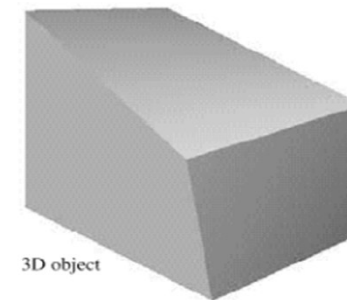
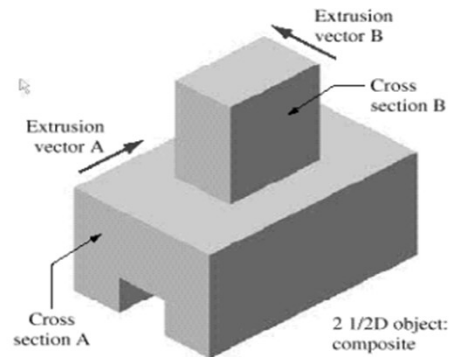
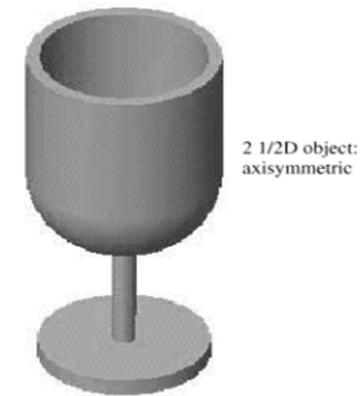
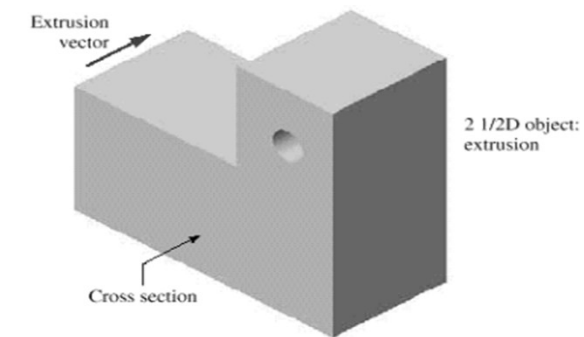
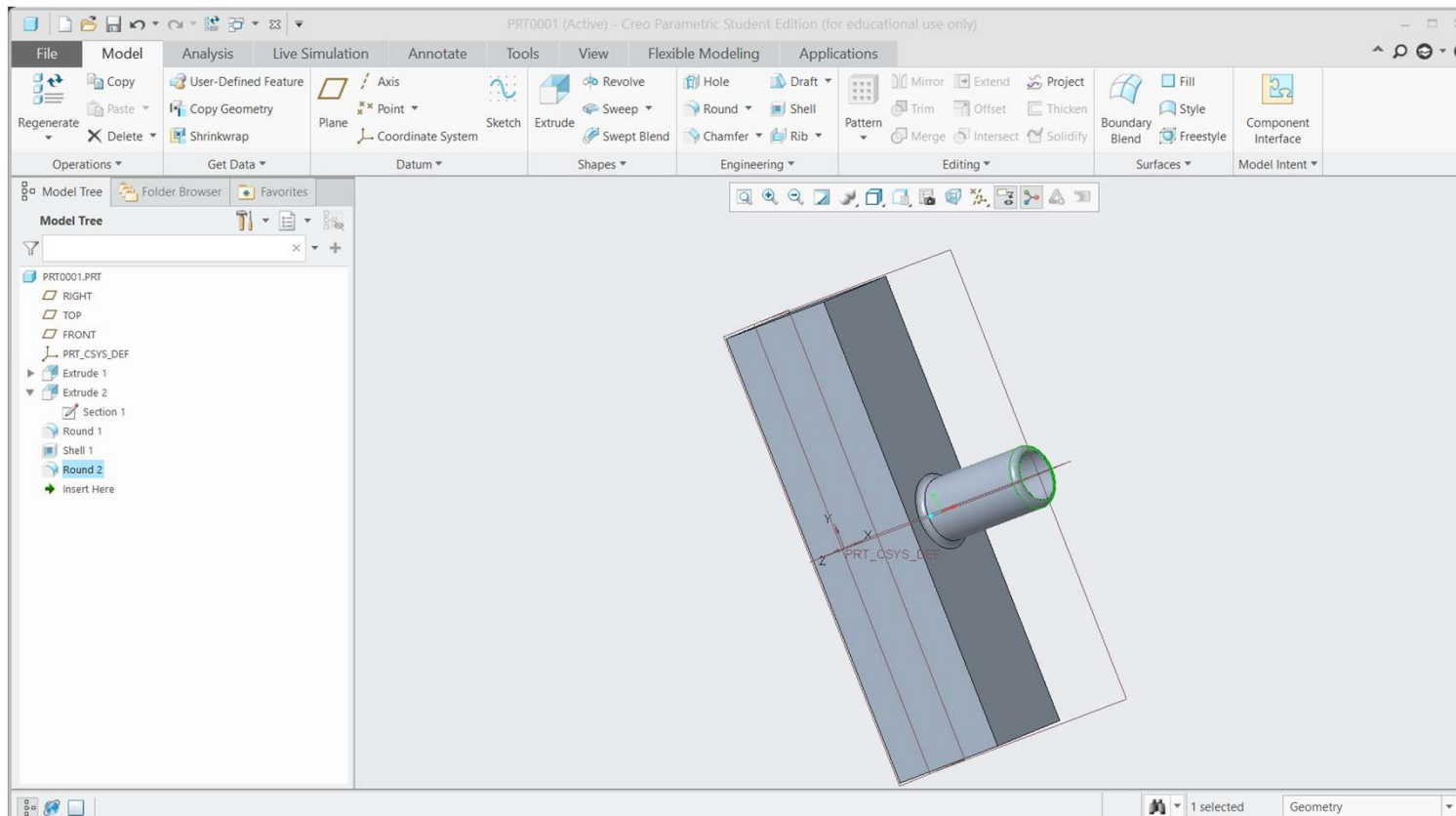


Figure 2.2 Types of geometric models.

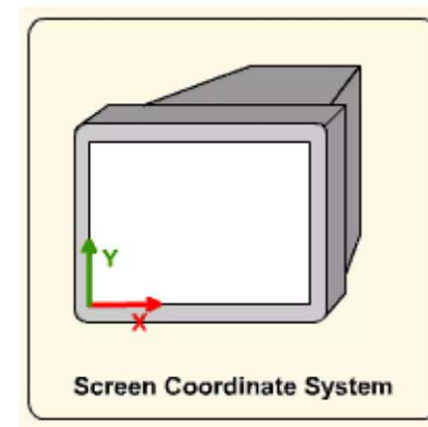
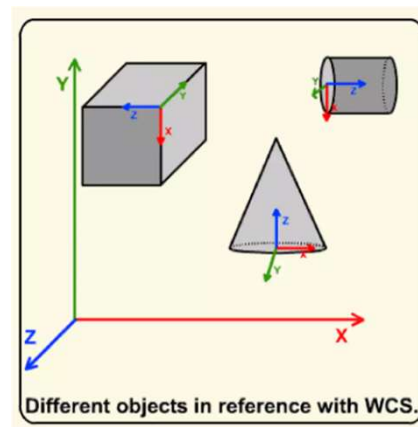
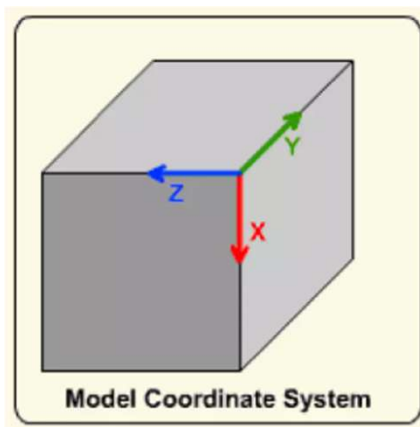
Typical GUI



Co-ordinate System

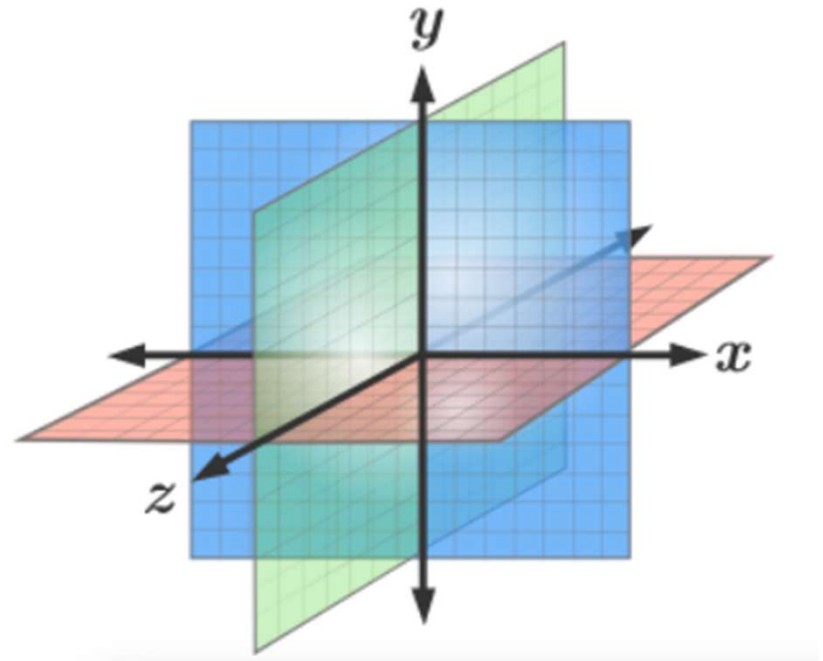


- MCS – Model Coordinate System
- WCS – Work Coordinate System
- SCS – Screen Coordinate System



Model Coordinate System (MCS)

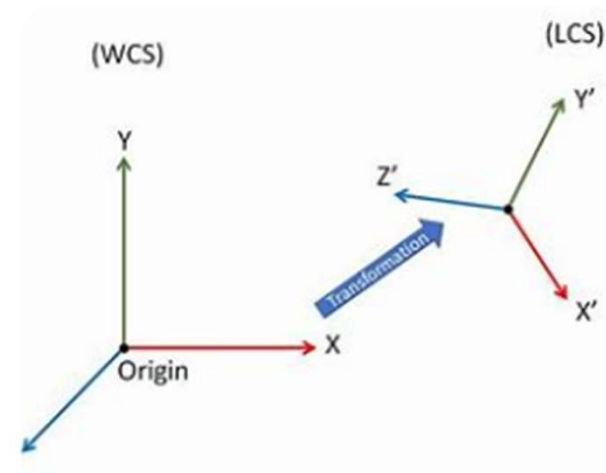
- Reference coordinate system of the model with respect to which position and orientation of the geometric model is defined.
- Default Cartesian coordinate systems
- Three default sketch planes and intersection point as MCS origin.



Working Coordinate System (WCS)



- Local CS typically used for modeling convenience/flexibility
- User defined CS
- Modeling dimensions can be specified with respect to LCS
- While saving, they are converted to MCS
- Only one WCS active at a time while modeling

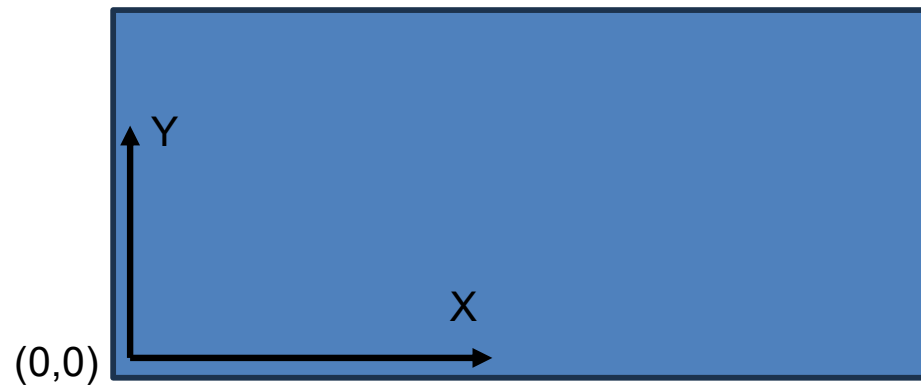


$$\mathbf{P}_m = \mathbf{T} \mathbf{P}_w$$

Screen Coordinate System (SCS)



- Device dependent CS
- Origin typically at left bottom
- MCS is mapped onto SCS for display purpose
- Used for selecting geometric entities of model
- A typical 1024x1024 pixel screen will have center coordinate as (512, 512)

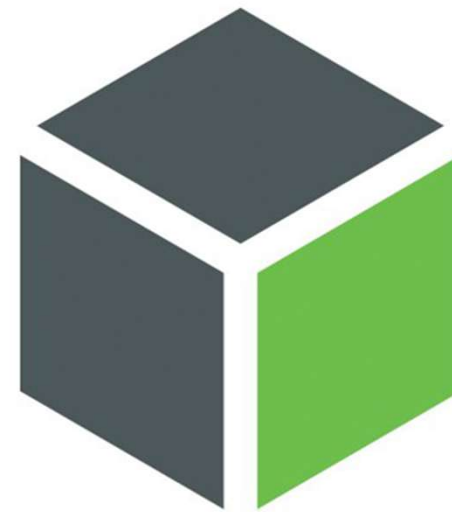


Download Creo Parametric



Student Version (Free):

<https://www.ptc.com/en/products/education/free-software/creo-college-download>



creo™ parametric



COMPUTER AIDED ANALYSIS AND DESIGN

Course (DE ZG622) - Lecture 3

Prof. Ruchi More Kotecha

Session Outline



- Overview of Computer Graphics
- Representation of Points
- Transformations and Matrices
- Transformations of Points and Lines
- Combined Transformations
- Homogenous Coordinates and Transformation Conventions
- Practice Problems

Overview of Computer Graphics

- Computer graphics is the use of computer to define, store, manipulate, interrogate, and present pictorial output.
- Computer displays that allow dynamic motion require that the picture/ image be redrawn at least 30 times per second in order to avoid flicker.
- Computer graphics can be divided into following stages:
 1. Representing pictures to be presented
 2. Preparing pictures for presentation
 3. Presenting previously prepared pictures
 4. Interacting with the picture





Stages of Computer Graphics

1. Representing pictures to be presented

- Pictures can be considered as a collection of lines, points and textual material.
- Points $\rightarrow$ single coordinate triplet (x_1, y_1, z_1)
- Lines $\rightarrow$ coordinates of its end points (x_1, y_1, z_1) and (x_2, y_2, z_2)
- Textual Material $\rightarrow$ collection of lines & points
- Curved Lines $\rightarrow$ approximation of short line segments

2. Preparing pictures for presentation

- Pictures ultimately consist of points. Points are basic building blocks of a graphic database.
- Database: File (array) in which coordinates of these points are stored prior to being used to present the picture.
- Ways to specify position of points:
 - i. Absolute coordinates
 - ii. Relative (incremental) coordinates
 - iii. Homogeneous coordinate system



Stages of Computer Graphics

3. Presenting previously prepared pictures

- Database (used to prepare picture for presentation) is never the same as Display File (used to present the picture).
- Database represents total picture while Display File represents only some part of the picture.
- Display File is created by transforming the Database (concatenation of matrices).
- Final step: Converting coordinates used in picture database (user coordinates) to those used by display device (display coordinates).

4. Interacting with the picture

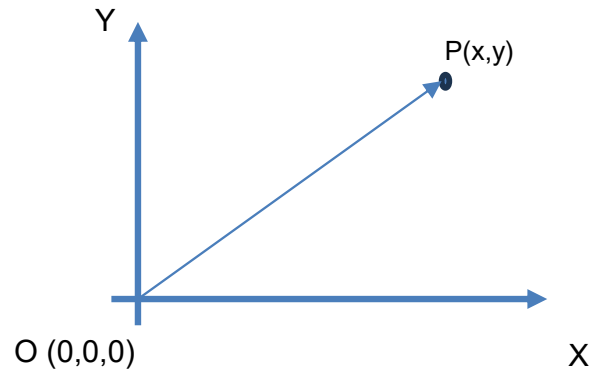
- Requires some sort of device to communicate with the program while its running.
- Types of devices: (i) alpha-numeric keyboard, (ii) joy stick, (iii) light pens, (iv) track balls, (v) mouse, (vi) control dials, (vii) function switches, (viii) analog tablets
- Classification of graphics devices:
 - i. Passive **vs.** Active
 - ii. Point Plotting **vs.** Line (Vector) Drawing
 - iii. Device accepts true 3D data **vs.** 3D data must be converted to 2D data



Representation of Points

- A point is a position specified with coordinate values in some reference frame.
- In mathematics, the origin of a Euclidean space is a special point, usually denoted by the letter O, used as a fixed point of reference for the geometry of the surrounding space.
- All points in the reference frame are given with respect to the origin.
- Any 2D object is represented in the form of points (vertex/ coordinates) and lines (vectors/ joining vertices).
- In 2D: a point (x, y) is represented as $[x \ y]$ or $\begin{bmatrix} x \\ y \end{bmatrix}$
- In 3D: a point (x, y, z) is represented as $[x \ y \ z]$ or $\begin{bmatrix} x \\ y \\ z \end{bmatrix}$
- A series of points, each of which is a position vector relative to a local coordinate system, may be stored in a computer as a matrix of numbers.
- The position of these points can be controlled by manipulating the matrix which defines the points.

Representation of a Point (2D)



- A point is a position specified with coordinate values in some reference frame.
- $P(x,y)$ represented in matrix form as

$$\begin{bmatrix} x & y \end{bmatrix} \text{ OR } \begin{bmatrix} x \\ y \end{bmatrix}$$



Transformation of Points

- Consider a 2×2 transformation matrix $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$

Initial coordinates $[x \ y]$ are transformed to $[x^* \ y^*]$ on application of the transformation matrix.

$$[x \ y] \begin{bmatrix} a & b \\ c & d \end{bmatrix} = [(ax + cy) \ (bx + dy)] = [x^* \ y^*]$$

where, $x^* = (ax + cy)$ and $y^* = (bx + dy)$

- Consider the case where $a = d = 1$ and $b = c = 0$

$$[x \ y] \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = [(1x + 0y) \ (0x + 1y)] = [x \ y] = [x^* \ y^*]$$

∴ No change in initial coordinates.

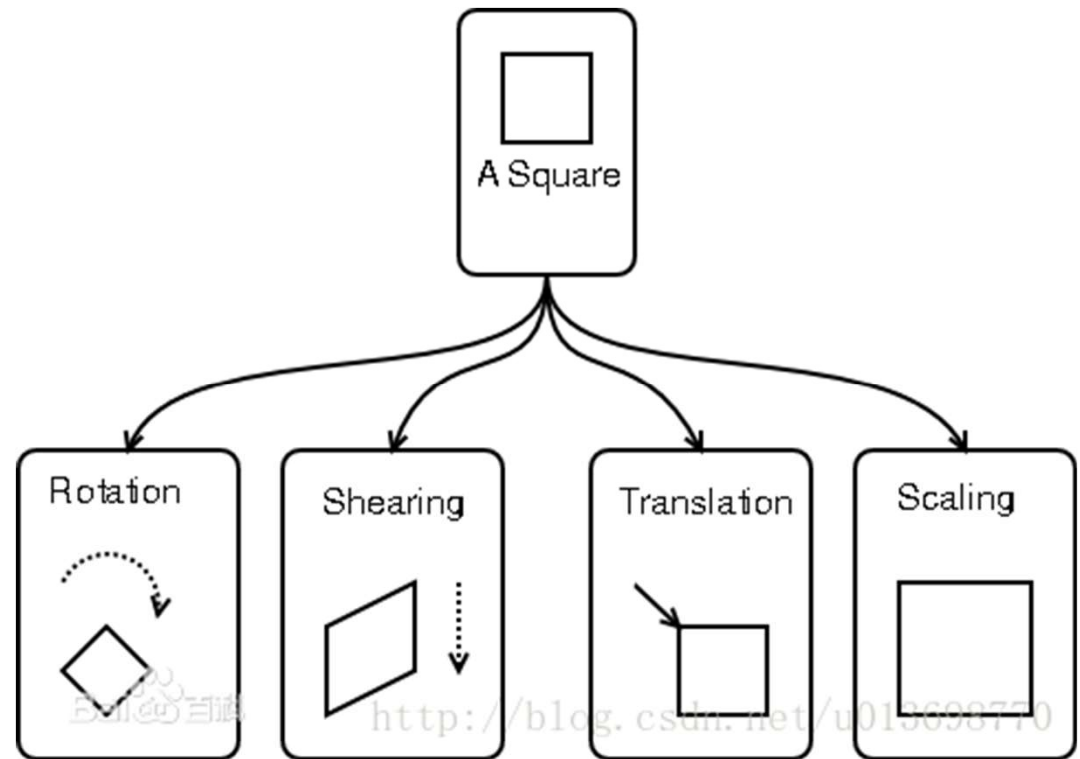
∴ Matrix $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ is called as an **identity matrix**.

Types of Transformations:

1. **Translation**
(Position)
2. **Scaling**
(Size)
3. **Rotation**
(Orientation)
4. **Shapes**
(Shear)

Transformations

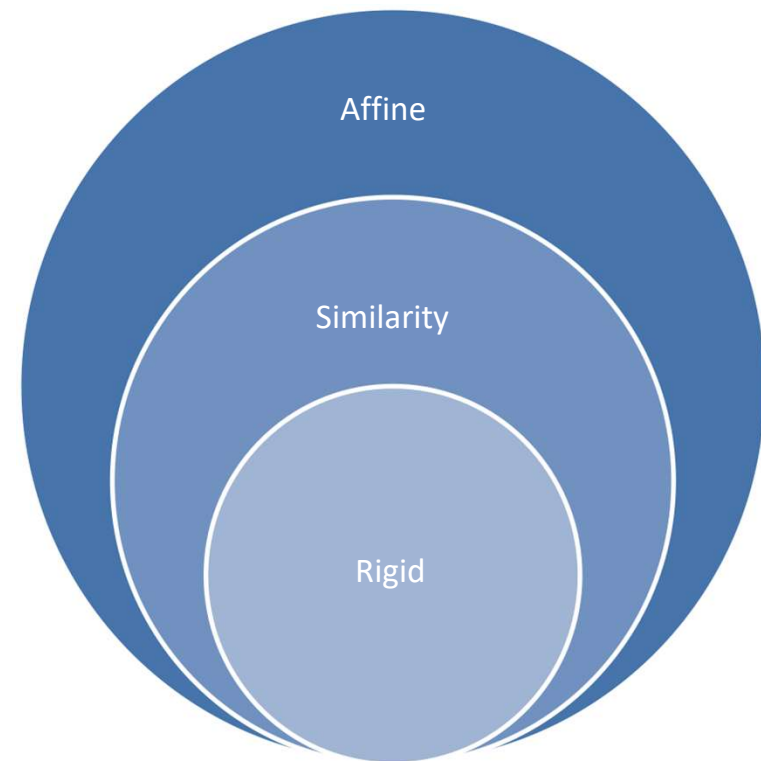
- **Rigid Transformations**
(Distance & Shape Preserving) :
Rotation + Translation
- **Similarity Transformations**
(Angle Preserving) :
Rotation + Translation + Uniform
Scale
- **Affine Transformations**
(Parallelism Preserving) :
Rotation + Translation + Uniform
Scale + Shear



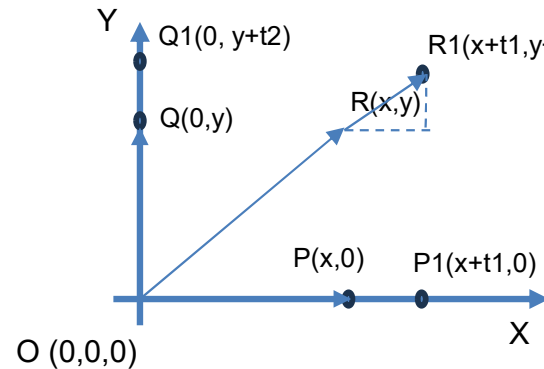
Types of Transformations



- **Rigid transformation (Preserve shape and distance):**
Translation, Rotation
- **Similarity transformation (Preserve angle):**
Translation, Rotation, Uniform Scale
- **Affine transformation (Preserve parallelism):**
Translation, Rotation, Scale, Shear



Translation of Point



$$P1 = [P] + [T]$$

- Point P(x,0) translated by distance t1 along X axis $\rightarrow P1(x+t1, 0)$
- Point Q(0, y) translated by distance t2 along Y axis $\rightarrow Q1(0, y+t2)$
- Point R(x, y) translated by distance t1 along X and by t2 along Y axis $\rightarrow R1(x+t1, y+t2)$

$$\begin{bmatrix} x & 0 \end{bmatrix} + \begin{bmatrix} t1 & 0 \end{bmatrix} = \begin{bmatrix} X+t1 & y \end{bmatrix}$$

$$\begin{bmatrix} 0 & y \end{bmatrix} + \begin{bmatrix} 0 & t2 \end{bmatrix} = \begin{bmatrix} X+t1 & y \end{bmatrix}$$

$$\begin{bmatrix} x & y \end{bmatrix} + \begin{bmatrix} t1 & t2 \end{bmatrix} = \begin{bmatrix} X+t1 & y+t2 \end{bmatrix}$$



Transformation of Points : Scaling

- Consider the case where $\mathbf{a} = \mathbf{d} = \mathbf{s}$ and $\mathbf{b} = \mathbf{c} = \mathbf{0}$

$$\begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix} = [(sx + 0y) \quad (0x + sy)] = [sx \quad sy] = [x^* \quad y^*]$$

∴ This produces a **scale change** since $x^* = sx$ and $y^* = sy$.

This matrix has an effect of stretching the original coordinate in x and y directions.

- Consider the case where $\mathbf{b} = \mathbf{c} = \mathbf{0}$

$$\begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} a & 0 \\ 0 & d \end{bmatrix} = [(ax + 0y) \quad (0x + dy)] = [ax \quad dy] = [x^* \quad y^*]$$

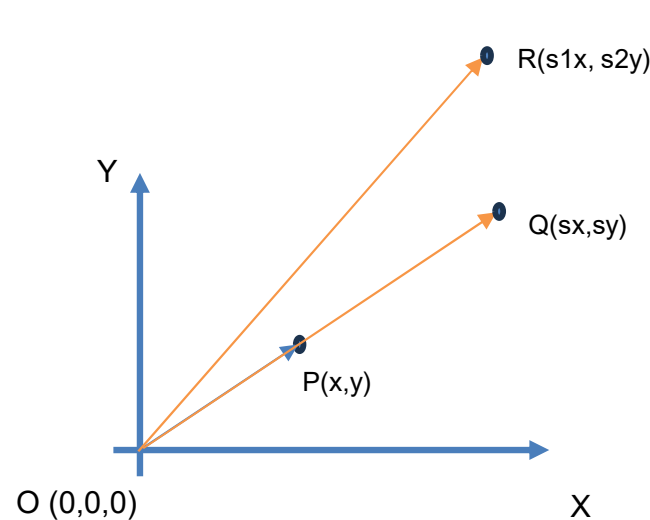
∴ This produces a **scale change** since $x^* = ax$ and $y^* = dy$

This matrix has an effect of stretching the original coordinate in both x & y directions. If stretching is unequal in both directions, it causes **distortion**.

Note:

- If $\mathbf{a} \neq \mathbf{d}$:
The stretching is not equal .
- If $\mathbf{a} = \mathbf{d} > 1$:
Pure enlargement occurs
- If $\mathbf{a} = \mathbf{d} < 1$:
Pure compression occurs

Scaling of Point (Position Vector)



$$\begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix} = \begin{bmatrix} (sx + 0y) & (0x + sy) \end{bmatrix}$$

$$= \begin{bmatrix} sx & sy \end{bmatrix} = \begin{bmatrix} x^* & y^* \end{bmatrix} - \text{(Symmetric scaling)}$$

$$\begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} s1 & 0 \\ 0 & s2 \end{bmatrix} = \begin{bmatrix} (s1x + 0y) & (0x + s2y) \end{bmatrix}$$

$$= \begin{bmatrix} s1x & s2y \end{bmatrix} = \begin{bmatrix} x^* & y^* \end{bmatrix} - \text{(Asymmetric scaling)}$$

$$P1 = [P] [S]$$



Transformation of Points : Reflection

- Consider the case where $\mathbf{a} = -1$, $\mathbf{d} = 1$ and $\mathbf{b} = \mathbf{c} = 0$

$$\begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} = [(-1x + 0y) \quad (0x + 1y)] = [-x \quad y] = [x^* \quad y^*]$$

∴ This produces a **reflection about y-axis**.

- Consider the case where $\mathbf{a} = 1$, $\mathbf{d} = -1$ and $\mathbf{b} = \mathbf{c} = 0$

$$\begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = [(1x + 0y) \quad (0x - 1y)] = [x \quad -y] = [x^* \quad y^*]$$

∴ This produces a **reflection about x-axis**.

- Consider the case where $\mathbf{a} = -1$, $\mathbf{d} = -1$ and $\mathbf{b} = \mathbf{c} = 0$

$$\begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} = [(-1x + 0y) \quad (0x - 1y)] = [-x \quad -y] = [x^* \quad y^*]$$

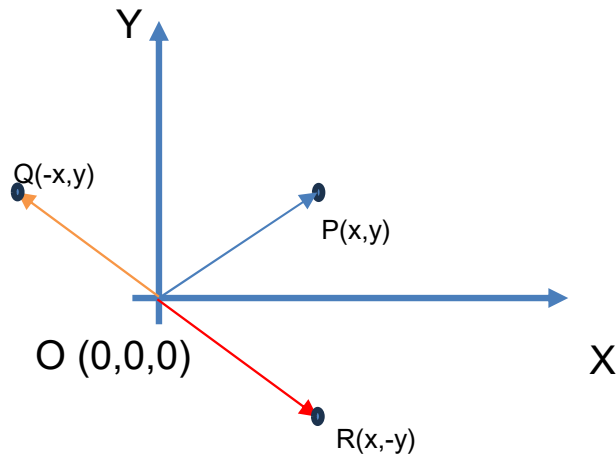
∴ This produces a **reflection about origin**.

Note:

It can be observed that :
when $b = c = 0$,
and either a or d or
both are negative,
then reflection
occurs.

However,
when $a = d = 0$,
and either b or c or
both are negative,
then reflection
does not occur .

Reflection of Point (Position Vector)



Reflection about Y axis

$$\begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} -x & y \end{bmatrix}$$

Reflection about X axis

$$\begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} x & -y \end{bmatrix}$$



Transformation of Points : Rotation

- Consider a 2×2 rotation matrix $\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$ where θ = angle of rotation

Initial coordinates $[x \ y]$ are transformed to $[x^* \ y^*]$ on application of the rotation matrix.

$$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = [(x\cos\theta - y\sin\theta) \ (x\sin\theta + y\cos\theta)]$$

where, $x^* = (x\cos\theta - y\sin\theta)$ and $y^* = (x\sin\theta + y\cos\theta)$

For $\theta = 90^\circ$

Rotation Matrix:

$$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

For $\theta = 180^\circ$

Rotation Matrix:

$$\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

For $\theta = 270^\circ$

Rotation Matrix:

$$\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

Note:

- If $\theta > 0$:
counter clockwise rotation
- If $\theta < 0$:
clockwise rotation
- It is necessary to remember order of matrix multiplication.

Derivation: <https://www.youtube.com/watch?v=EZufilwwqFA>



Transformation of Points : Shear

- Consider the case where $\mathbf{a = d = 1}$, $\mathbf{c = 0}$ and $\mathbf{b = k}$

$$\begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = [(1x + ky) \quad (0x + 1y)] = [x + ky \quad y] = [x^* \quad y^*]$$

The y-coordinate of the point remains unchanged while the x-coordinate depends linearly on the original coordinates.

∴ This produces a **horizontal shear** since $x^* = x + ky$

- Consider the case where $\mathbf{a = d = 1}$, $\mathbf{b = 0}$ and $\mathbf{c = k}$

$$\begin{bmatrix} 1 & 0 \\ k & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = [(1x + 0y) \quad (kx + 1y)] = [x \quad kx + y] = [x^* \quad y^*]$$

The x-coordinate of the point remains unchanged while the y-coordinate depends linearly on the original coordinates.

∴ This produces a **vertical shear** since $y^* = kx + y$

Note:

- A shear transformation has the effect of slanting the shape.





Transformation of Origin, Straight Lines:

- Consider the effect of a 2×2 transformation matrix on the origin

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} (ax + by) & (cx + dy) \end{bmatrix} = \begin{bmatrix} x^* & y^* \end{bmatrix}$$

For the origin, $\mathbf{x} = \mathbf{0}$ and $\mathbf{y} = \mathbf{0}$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} (a0 + b0) & (c0 + d0) \end{bmatrix} = \begin{bmatrix} 0 & 0 \end{bmatrix} = \begin{bmatrix} x^* & y^* \end{bmatrix}$$

We can see that the **origin is invariant** under a 2×2 transformation.

Transformation of Straight Lines:

- A straight line can be defined by two position vectors which specify the coordinates of its end points.
- The position and orientation of the line joining these two points can be changed by operating on these two position vectors.
- Transformations are applied to each and every point defining the line.

Note:

- The rigidity of transformation of straight lines holds true for each and every point lying on it.
- When a 2×2 matrix is used to transform a pair of parallel lines, the result is a second pair of parallel lines.

Combined (Composite) Transformations



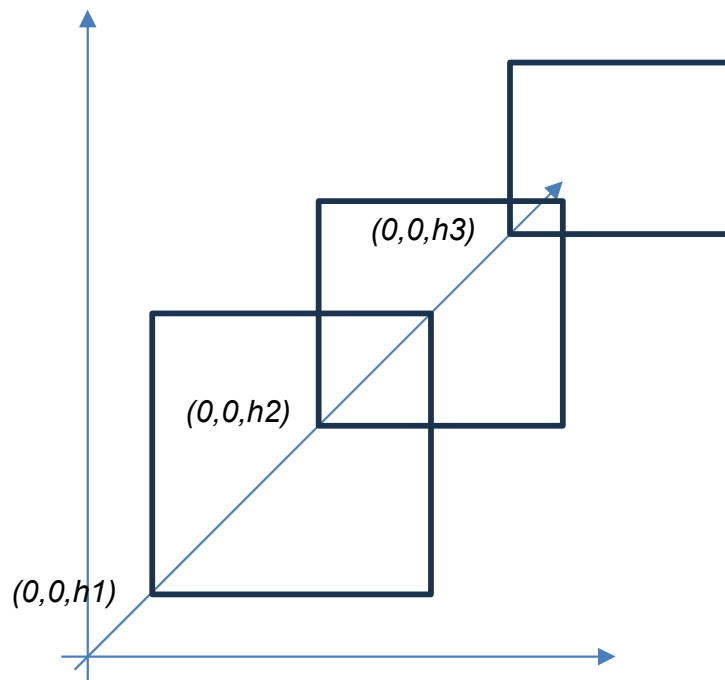
- A sequence of transformations can be collapsed into a single matrix
- This is also called concatenation of matrices

$$\begin{bmatrix} A \end{bmatrix} \begin{bmatrix} B \end{bmatrix} \begin{bmatrix} C \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} D \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

Note:

If a matrix is represented in column form, then the composite transformation is performed by multiplying matrix in order from right to left side.

Homogeneous Coordinate System

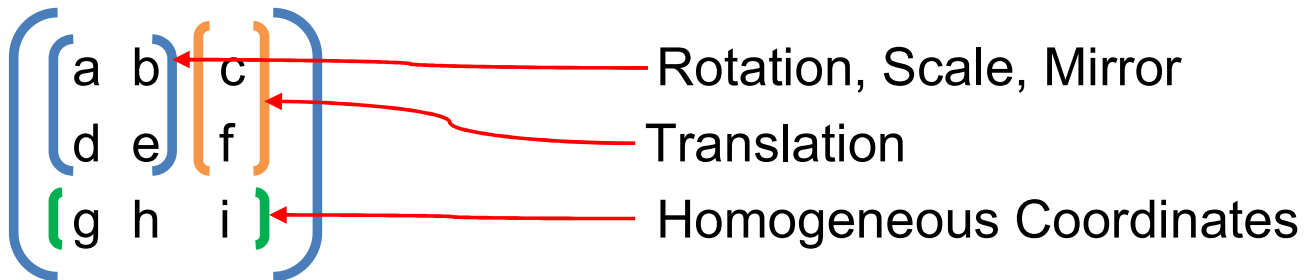


- n dimensional space is mapped to $(n+1)$ dimensional space
- So $P(x, y, z)$ is represented as $P(x, y, z, h)$
- For 2D case $P(x, y) = P(x, y, h)$ (h is typically 1)

Concatenated Matrix Representation



$$[R][S][T][R][P] = [M][P] = [P^*]$$





Transformation of Points : Translation

- Translation Transformation in the addition form cannot be combined with other transformations. Hence, in order to overcome this issue, we convert the 2×2 translation matrix into a 3×3 translation matrix with the use of homogeneous coordinates.
- Matrix Translation by Matrix Multiplication (using a 3×3 matrix)

$$\begin{bmatrix} a & b & u \\ c & d & v \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

For a translation formula, we set the a, b, c, d values to a unit matrix, and set the u, v elements to contain the translation values. When we apply this matrix to a position vector we do indeed get a translation:

$$\begin{bmatrix} 1 & 0 & u \\ 0 & 1 & v \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} x + u \\ y + v \\ 1 \end{bmatrix}$$

- We extend our transformation matrix by adding an extra column to the right, containing the transformation values u and v. To keep the matrix square, we add an extra row of 0, 0, 1.
- We also need to extend our position vector to be 3 elements long. We do this by adding an extra row element that is always equal to 1.

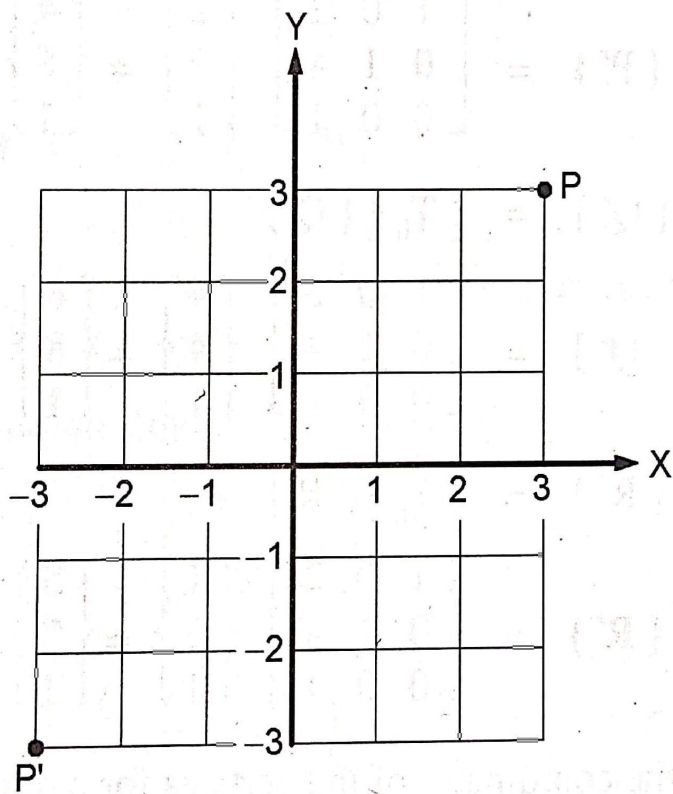
Practice Problems



Problem 1:

If a point P having coordinates (3, 3) is mirrored about X and Y axis, determine its new coordinates.

Answer for Practice Problem 1



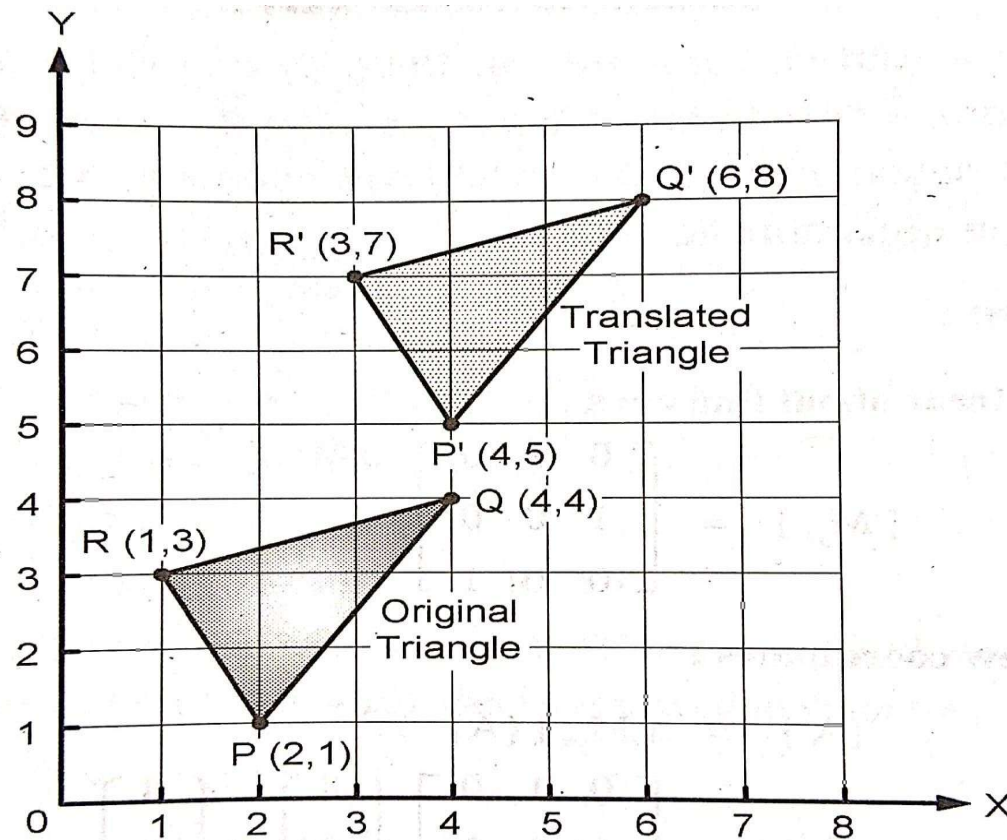
Practice Problems



Problem 2:

A triangle PQR with vertices P (2, 1), Q (4, 4) and R (1, 3) is to be translated by 2 units in X-direction and 4 units in Y-direction. Using the homogeneous coordinate system, determine the coordinates of the vertices for a translated triangle

Answer for Practice Problem 2



Practice Problems



Problem 3:

An object is to be rotated about point A $(-10, -10)$ by 90° in CCW direction. Find the concatenated transformation matrix.

Answer for Practice Problem 3



$$[CT] = \begin{bmatrix} 0 & -1 & -20 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

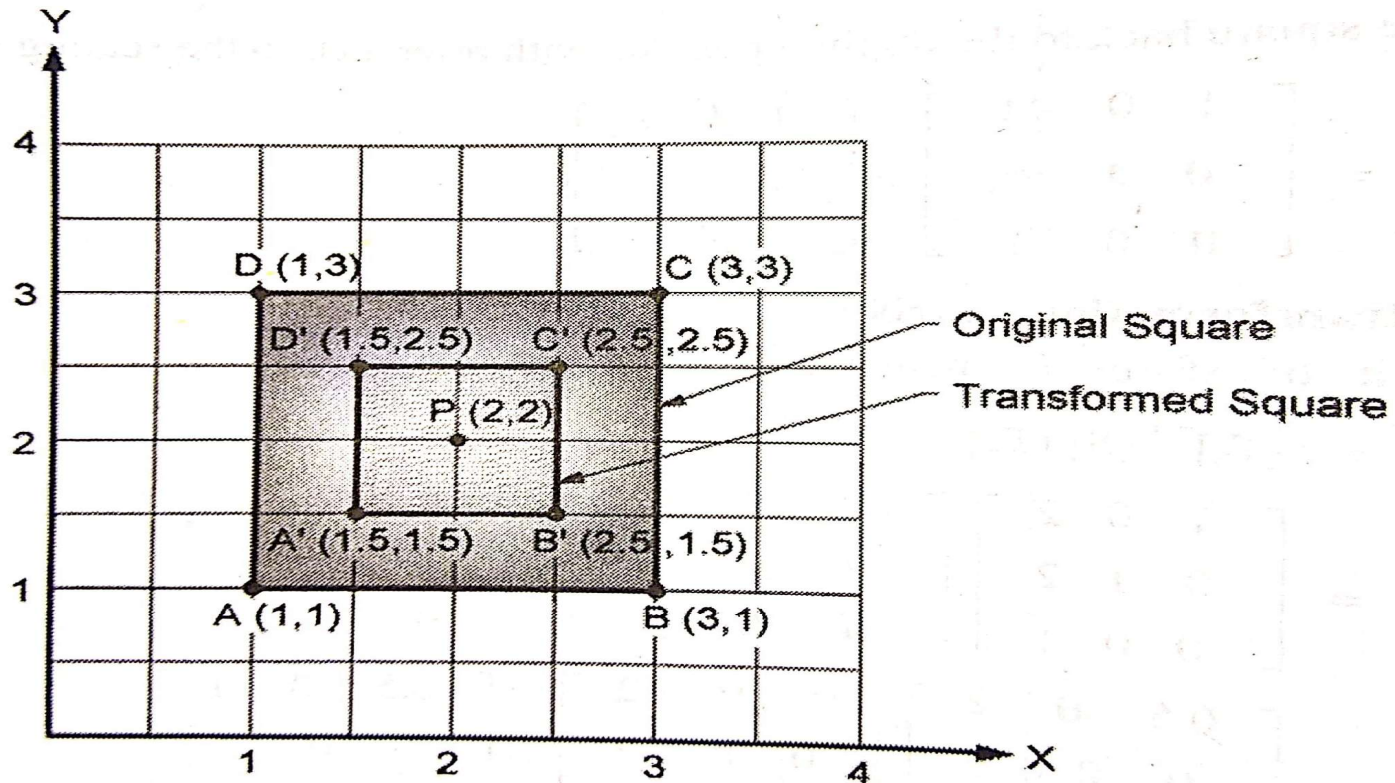
Practice Problems



Problem 4:

The square ABCD with vertices A (1,1), B (3, 1), C (3, 3) and D (1, 3) is to be transformed to half its size, still retaining the square at the same position. Determine the transformation matrix and the coordinates of the transformed square.

Answer for Practice Problem 4



Practice Problems

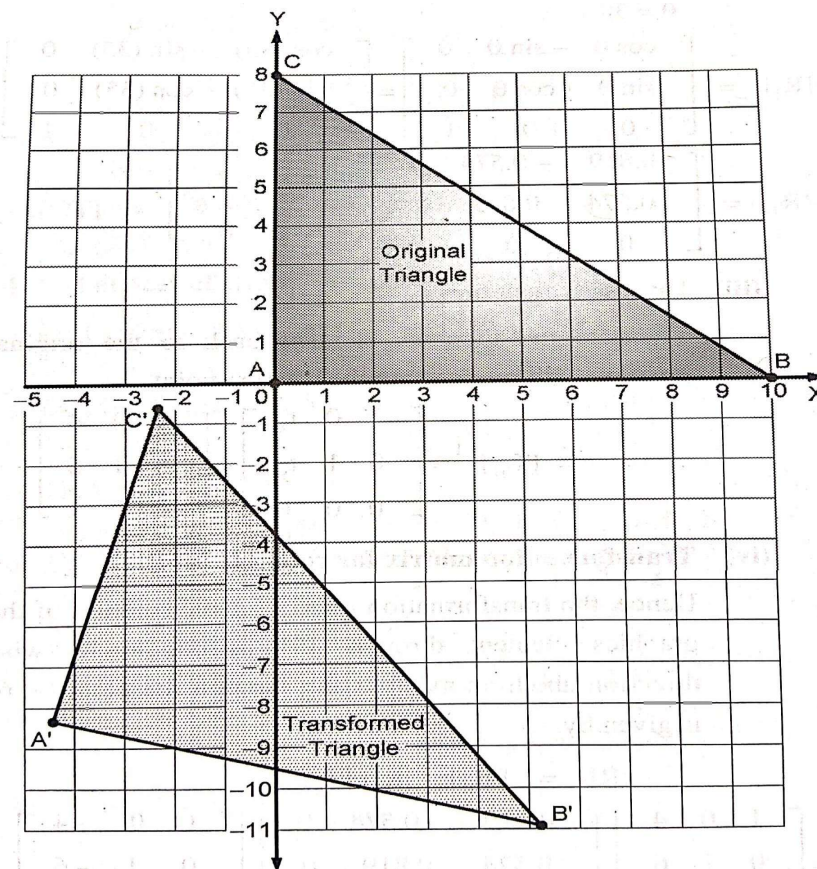


Problem 5:

What is the effect of the following transformations on a triangle having coordinates A (0, 0), B (10, 0) and C (0, 8)?

- (i) Rotation through 45° in anticlockwise direction
- (ii) Translation through +5 and -8 units along X and Y directions respectively
- (iii) Rotation through 60° in clockwise direction

Answer for Practice Problem 5





COMPUTER AIDED ANALYSIS AND DESIGN

Course (DE ZG622) - Lecture 4

Prof. Ruchi More Kotecha

Session Outline



- 3D Transformation basics and comparison
- General 4×4 Transformation Matrix
- Transformations and Matrices
- Orthographic & Isometric Projections
- Practice Problems

3D Transformations



- 3D transformation is an extension of 2D transformation in space.

2D Transformations

Transformations take place in a plane

Point is described by (x, y) coordinates in XY plane.

Geometry system is described by the intersection of two planes which are perpendicular to each other.

Coordinate system is divided into 4 quadrants.

3D Transformations

Transformations take place in space

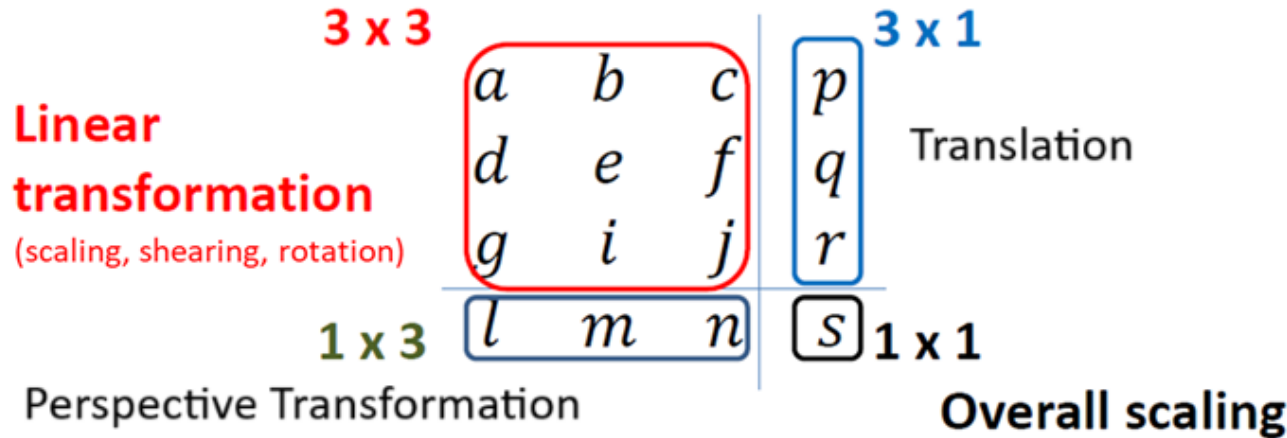
Point is described by (x, y, z) coordinates in space.

Geometry system is described by the intersection of three planes which are perpendicular to each other.

Coordinate system is divided into eight octants.

General 4x4 Transformation Matrix

- 3D transformation is an extension of 2D transformation in space.
- A point in 3D space $[x \ y \ z]$ is represented by a 4D position vector $[x \ y \ z \ 1]$ or $[X \ Y \ Z \ H]$.
- The general transformation vector is given by:



3D Transformation : Translation

- $x' = x + t_x$
 $y' = y + t_y$
 $z' = z + t_z$

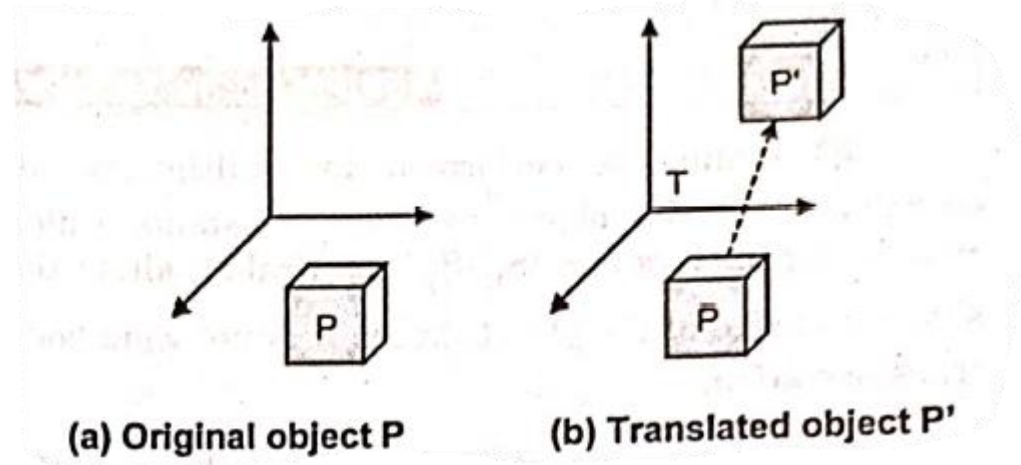
- $P' = [T][P]$

- Translation Matrix:

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & t_x \\ 0 & 1 & 0 & t_y \\ 0 & 0 & 1 & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

- Inverse Translation Matrix:

$$[T^{-1}] = \begin{bmatrix} 1 & 0 & 0 & -t_x \\ 0 & 1 & 0 & -t_y \\ 0 & 0 & 1 & -t_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$



3D Transformation : Scaling

- **Scaling with respect to Origin:**

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} S_x & 0 & 0 & 0 \\ 0 & S_y & 0 & 0 \\ 0 & 0 & S_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

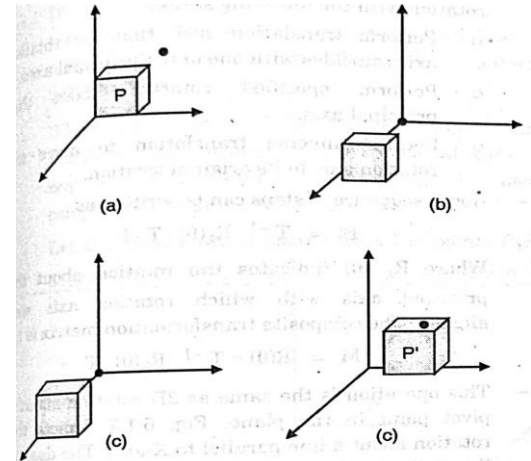
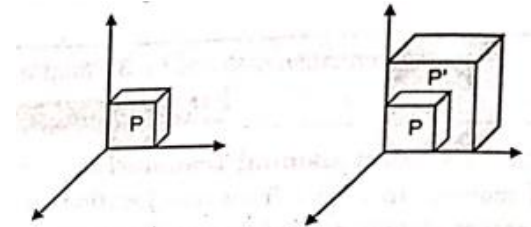
- If all 3 parameters are the same, then it is called **uniform scaling**. Otherwise, it is called **non-uniform scaling**.

- **Scaling with respect to a Reference Point:**

Reference Point = $[x_r \ y_r \ z_r]$

$$P' = [T^{-1}][S][T][P]$$

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} S_x & 0 & 0 & (1 - S_x)x_r \\ 0 & S_y & 0 & (1 - S_y)y_r \\ 0 & 0 & S_z & (1 - S_z)z_r \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$



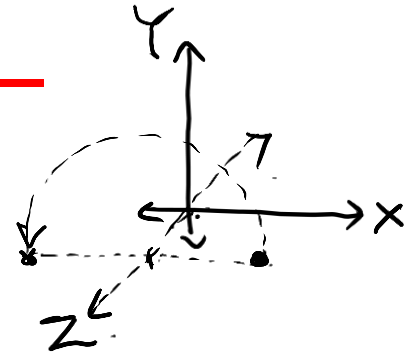
3D Rotation

$$\rightarrow \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



3D rotation about any axis:

- 1) angle of rotation
- 2) axis of rotation.



Rotation @ Z-axis:

$$x' = \underline{x \cos \theta} - \underline{y \sin \theta}$$

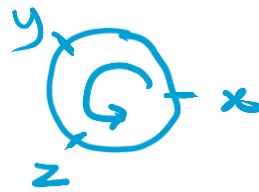
$$y' = \underline{x \sin \theta} + \underline{y \cos \theta}$$

$$\underline{z'} = \underline{z}$$

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta & 0 & 0 \\ \sin \theta & \cos \theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Cyclic Permutation:

3D Rotation



Cyclic Permutation: $x \rightarrow y \rightarrow \underline{z} \rightarrow \textcircled{x}$.

Rotation @ x -axis:

$$y' = y \cos \theta - z \sin \theta \leftarrow$$

$$z' = y \sin \theta + z \cos \theta \leftarrow$$

$$\underline{x'} = x$$

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta & 0 \\ 0 & \sin \theta & \cos \theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Rotation @ z -axis:

$$\rightarrow x' = \underline{x \cos \theta} - \underline{y \sin \theta}$$

$$y' = \underline{x \sin \theta} + y \underline{\cos \theta}$$

$$z' = \underline{z}$$

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta & 0 & 0 \\ \sin \theta & \cos \theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Rotation @ y -axis:

$$z' = \underline{z \cos \theta} - \underline{x \sin \theta} \leftarrow$$

$$x' = \underline{z \sin \theta} + x \underline{\cos \theta} \leftarrow$$

$$y' = y \leftarrow$$

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos \theta & 0 & \sin \theta & 0 \\ 0 & 1 & 0 & 0 \\ -\sin \theta & 0 & \cos \theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

3D Transformation : Rotation

➤ Rotation about X-axis:

→ $[R_x(\theta)] = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta & 0 \\ 0 & \sin\theta & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

$R_{x,\theta}$

➤ Rotation about Y-axis:

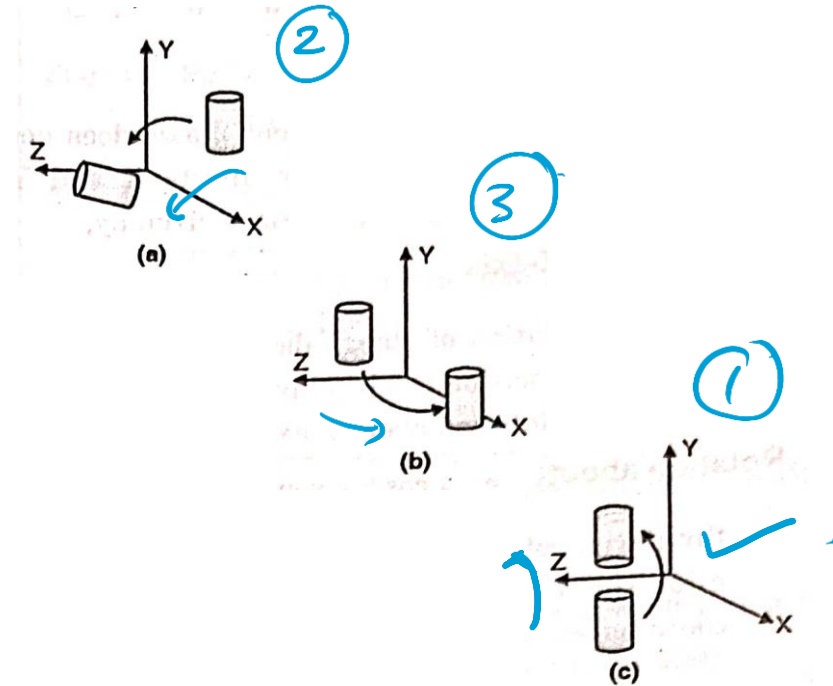
→ $[R_y(\theta)] = \begin{bmatrix} \cos\theta & 0 & \sin\theta & 0 \\ 0 & 1 & 0 & 0 \\ -\sin\theta & 0 & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

$R_{y,\theta}$

➤ Rotation about Z-axis:

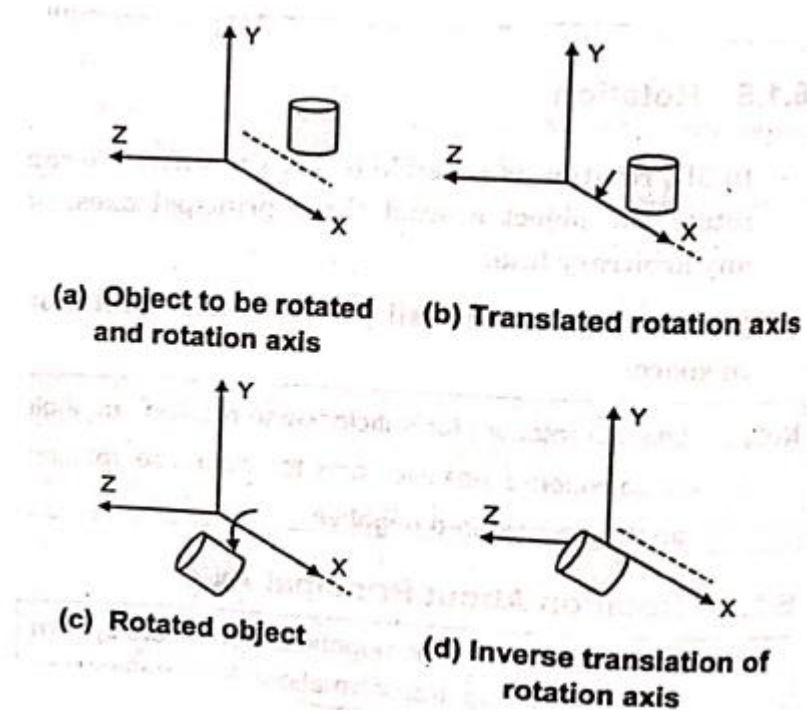
→ $[R_z(\theta)] = \begin{bmatrix} \cos\theta & -\sin\theta & 0 & 0 \\ \sin\theta & \cos\theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

$R_{z,\theta}$



3D Transformation : Rotation

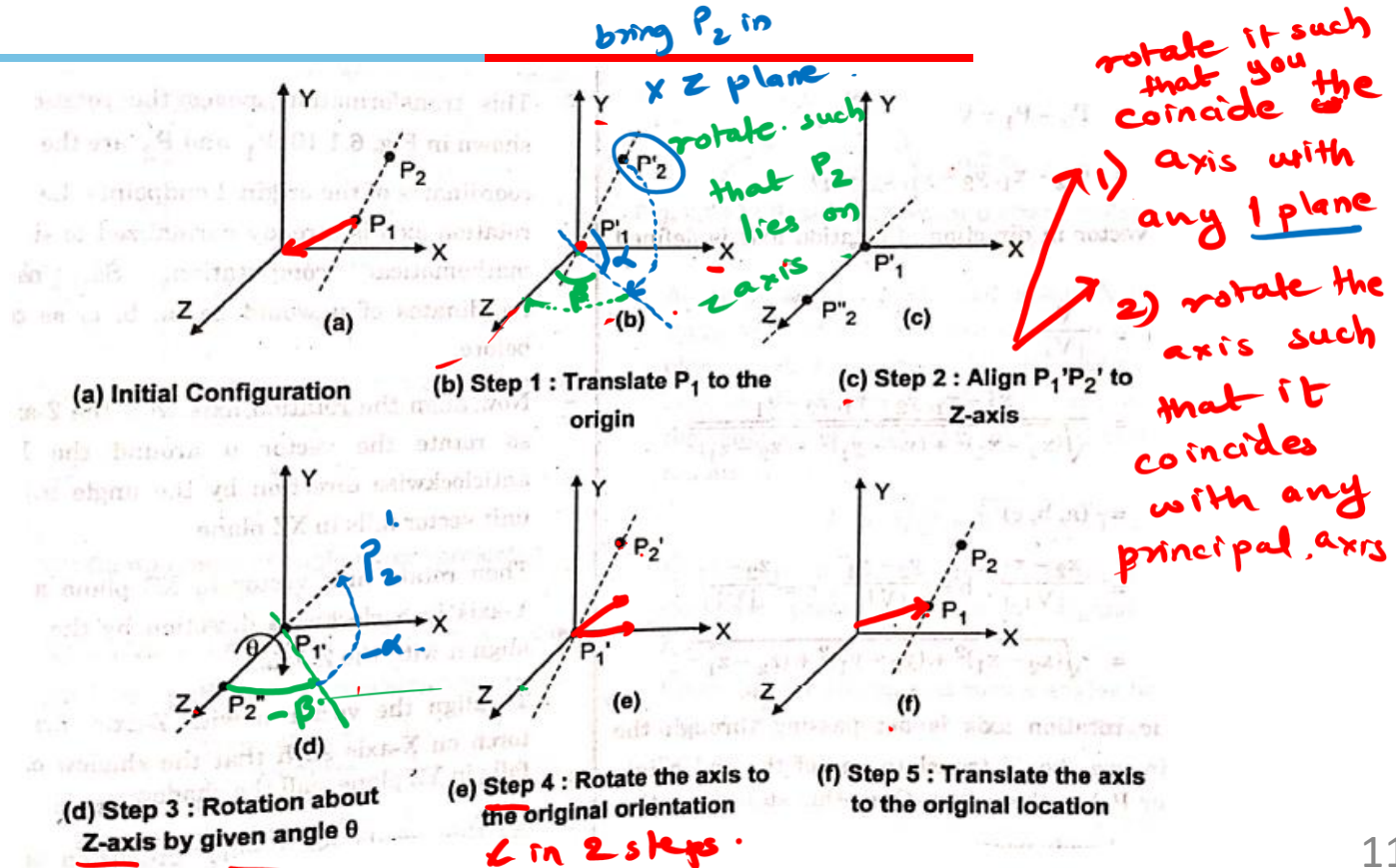
- **Rotation about line parallel to principal axis:**
- **1.** Perform translation such that rotation axis coincides with the principal axis to which it is originally parallel.
- **2.** Perform specified rotation about that principal axis
- **3.** Perform inverse translation to move the rotation axis back to its original location.
- $P' = [T^{-1}][R_{\lambda}(\Theta)][T][P]$



3D Transformation : Rotation



- Rotation about any arbitrary line:



3D Transformation : Reflection

Reflection about an axis is equivalent to rotation by 180° about that axis.

Reflection about planes $X=0$, $Y=0$ and $Z=0$:

➤ $\text{Ref}_{x=0} = \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$
In Y-Z plane

➤ $\text{Ref}_{z=0} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$
In X-Y plane

➤ $\text{Ref}_{y=0} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$
In XZ plane

➤ $[P'] = [\text{Ref}] [P]$

3D Transformation : Shear

➤ $SH_x = \begin{bmatrix} 1 & 0 & 0 & 0 \\ a & 1 & 0 & 0 \\ b & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

➤ $SH = \begin{bmatrix} 1 & c & e & 0 \\ a & 1 & f & 0 \\ b & d & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

The off diagonal terms in upper left 3x3 matrix of the generalized 4x4 transformation matrix produces shear in the 3 respective directions.

➤ $SH_y = \begin{bmatrix} 1 & c & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & d & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

➤ $[P'] = [SH] [P]$

➤ $\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & c & e & 0 \\ a & 1 & f & 0 \\ b & d & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$

➤ $P' = [(x + cy + ez) \quad (ax + y + fz) \quad (bx + dy + z)]$

➤ $SH_z = \begin{bmatrix} 1 & 0 & e & 0 \\ 0 & 1 & f & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

Orthographic Projections

- When the point is projected on any plane using a perpendicular projection, the third coordinate becomes zero. Any point on the XY plane will be (x, y, 0) and any point on X-axis will be (x, 0, 0).
- Projections onto one of the coordinate planes $x=0$, $y=0$ or $z=0$

$x=0$

$$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$y=0$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$z=0$

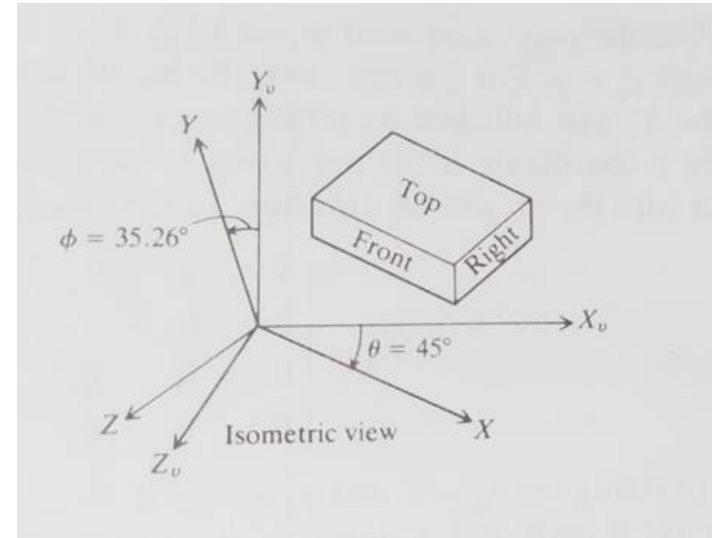
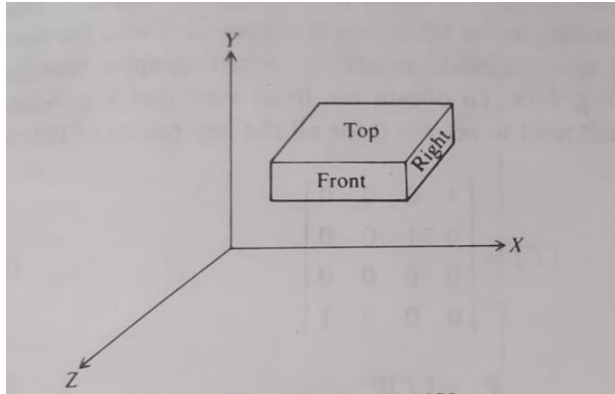
$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

<https://www.youtube.com/watch?v=Rz4WWtDX2Kc>

Isometric Projections



- Isometric projection is obtained by rotating object by 45 degrees about Y axis and then by rotating about X axis by 35.26 degrees (conveniently taken as 30 degrees).



<https://www.youtube.com/watch?v=8j7l1OWhMIE>

Isometric Projections

➤ Resulting Transformation

θ = Angle rotation about X axis

ϕ = Angle of rotation about Y axis

$$P_r = [T_x][T_y]P$$

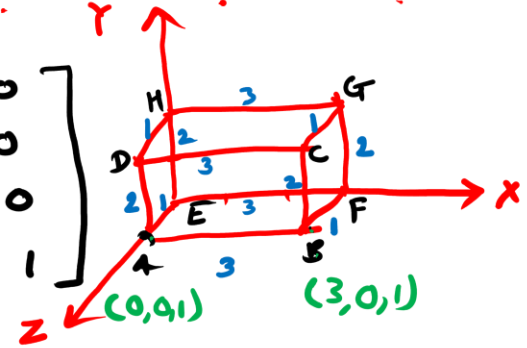
$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \phi & -\sin \phi & 0 \\ 0 & \sin \phi & \cos \phi & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos \theta & 0 & \sin \theta & 0 \\ 0 & 1 & 0 & 0 \\ -\sin \theta & 0 & \cos \theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

Practice Problems

Practice Problem :

A rectangular parallelepiped has its length as 3 units on X axis, 2 units on Y axis and 1 unit on the Z axis. Perform rotation by 90° clockwise about X axis.

$$[R] = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos(-90) & -\sin(-90) & 0 \\ 0 & \sin(-90) & \cos(-90) & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



$$[P'] = [R] \cdot [P]$$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} A & B & C & D & E & F & G & H \\ 0 & 3 & 3 & 0 & 0 & 3 & 3 & 0 \\ 0 & 0 & 2 & 2 & 0 & 0 & 2 & 2 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} A' & B' & C' & D' & E' & F' & G' & H' \\ 0 & 3 & 3 & 0 & 0 & 3 & 3 & 0 \\ 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & -2 & -2 & 0 & 0 & -2 & -2 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

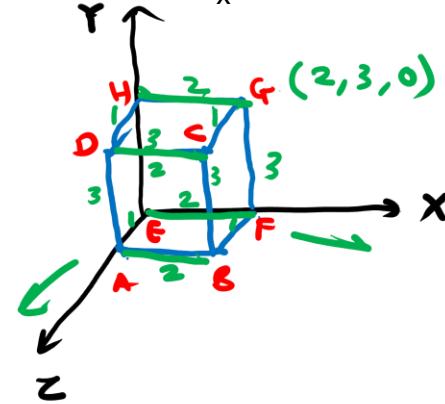
Practice Problem :

Given is a rectangular parallelopiped which has its length as 1 unit on the Z axis, 2 units on X axis and 3 units on Y axis. What is the effect of scaling when scaling factors are $S_x = 1/2$, $S_y = 1/3$ and $S_z = 1$.

$$[S] = \begin{bmatrix} s_x & 0 & 0 & 0 \\ 0 & s_y & 0 & 0 \\ 0 & 0 & s_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$[P'] = [S] \cdot [P]$$

$$= \begin{bmatrix} 1/2 & 0 & 0 & 0 \\ 0 & 1/3 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} A & B & C & D & E & F & G & H \\ 0 & 2 & 2 & 0 & 0 & 2 & 2 & 0 \\ 0 & 0 & 3 & 3 & 0 & 0 & 3 & 3 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

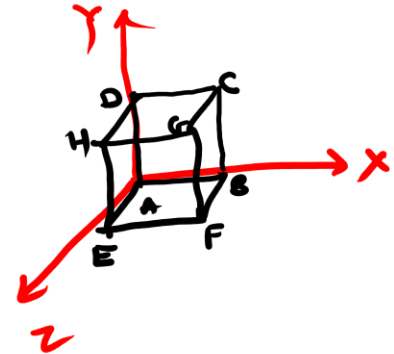


Practice Problem :

A unit cube is translated by 2 units in X direction, 3 units in Y direction and 4 units in Z direction. Find the translated coordinates of the cube.

$$[T] = \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$[P] = \begin{bmatrix} \text{A} & \text{B} & \text{C} & \text{D} & \text{E} & \text{F} & \text{G} & \text{H} \\ 0 & 1 & 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

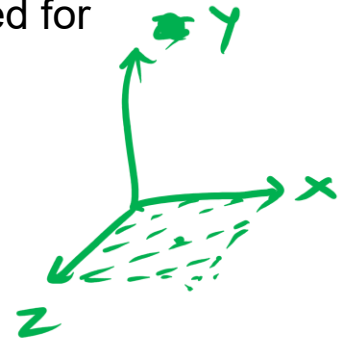


$$[P'] = [T] \cdot [P]$$

Practice Problem :

A line segment defined by endpoints ^A(10,10,20) and ^B(20,50,60) needs to be orthographically projected on to XZ plane. Find the transformation matrix used for this orthographic projection and coordinates of projected line segment

projected onto XZ plane $\rightarrow$ Y coordinate is 0.



Point B:

$$\begin{bmatrix} x'_B \\ y'_B \\ z'_B \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \times \begin{bmatrix} 20 \\ 50 \\ 60 \\ 1 \end{bmatrix}$$

Point-A

$$\begin{bmatrix} x'_A \\ y'_A \\ z'_A \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \times \begin{bmatrix} 10 \\ 10 \\ 20 \\ 1 \end{bmatrix}$$

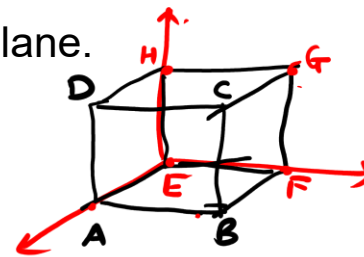
Practice Problem :

Find the orthographic projection of a unit cube onto the $x=0$, $y=0$ & $z=0$ plane.

Matrix for projection when $x=0$ [yz plane]

$$[M] = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$P = \begin{bmatrix} 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$



~~$$[P'] = [M][P]$$~~

$$[P'] = [P][M]$$

Practice Problem :

$$P_1' [10, 8, 0] \quad P_2' [10, 10, 0]$$

A line is defined by end points P1 (10,8,9) and P2 (10,10,10). It needs to be orthographically projected onto the XY plane. Write the coordinates of the projected line segment.

$$z=0$$

$$[M] = \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & 0 & \\ & & & 1 \end{bmatrix}$$

$$A : [M] \begin{bmatrix} 10 \\ 8 \\ 9 \\ 1 \end{bmatrix} = \begin{bmatrix} 10 \\ 8 \\ 0 \\ 1 \end{bmatrix}$$

$$B : \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 10 \\ 10 \\ 10 \\ 1 \end{bmatrix} = \begin{bmatrix} 10 \\ 10 \\ 0 \\ 1 \end{bmatrix}$$

Practice Problem 1:



Rotate a triangle ABC with vertices A (2, 2, 2), B (3, 4, 7) and C (8, 9, 12) about Y axis in anticlockwise direction by angle 90° .

Solution for Practice Problem 1:



Sol: The rotation angle is 90° and rotation is to be performed in an anticlockwise direction, so $\theta = +90^\circ$. Rotation is to be performed about the Y-axis, so transformed co-ordinates are computed as,

$$P' = R_y(\theta) \cdot P = \begin{bmatrix} \cos 90^\circ & 0 & \sin 90^\circ & 0 \\ 0 & 1 & 0 & 0 \\ -\sin 90^\circ & 0 & \cos 90^\circ & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 3 & 8 \\ 2 & 4 & 9 \\ 2 & 7 & 12 \\ 1 & 1 & 1 \end{bmatrix}$$
$$P' = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 3 & 8 \\ 2 & 4 & 9 \\ 2 & 7 & 12 \\ 1 & 1 & 1 \end{bmatrix}$$

Solution for Practice Problem 1:



$$= \begin{bmatrix} 2 & 7 & 12 \\ 2 & 4 & 9 \\ -2 & -3 & -8 \end{bmatrix}$$

Rotated vertices are $A'(2, 2, -2)$, $B'(7, 4, -3)$ and $C'(12, 9, -8)$

Original Coordinates	Transformed coordinates
$A(2, 2, 2)$	$A'(2, 2, -2)$
$B(3, 4, 7)$	$B'(7, 4, -3)$
$C(8, 9, 12)$	$C'(12, 9, -8)$

Practice Problem 2:



Find the transformation matrix for rotation about a line parallel to the X-axis and passing through the reference point P (x_r , y_r , z_r).

Solution for Practice Problem 2:



Mentioned composite transformation is achieved by performing the sequence of the following steps:

1. Translation by $T(-x_h, -y_h, -z_h)$
2. Rotation about x -axis by angle θ .
3. Inverse translation.

$$\therefore M = T^{-1} \cdot R_x(\theta) \cdot T$$
$$= \begin{bmatrix} 1 & 0 & 0 & x_h \\ 0 & 1 & 0 & y_h \\ 0 & 0 & 1 & z_h \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta & 0 \\ 0 & \sin\theta & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & -x_h \\ 0 & 1 & 0 & -y_h \\ 0 & 0 & 1 & -z_h \\ 0 & 0 & 0 & 1 \end{bmatrix}$$
$$= \begin{bmatrix} 1 & 0 & 0 & x_h \\ 0 & 1 & 0 & y_h \\ 0 & 0 & 1 & z_h \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & -x_h \\ 0 & \cos\theta & -\sin\theta & -y_h \cos\theta + z_h \sin\theta \\ 0 & \sin\theta & \cos\theta & y_h \sin\theta - z_h \cos\theta \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Solution for Practice Problem 2:



$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta & -y_r \cos \theta + z_r \sin \theta + y_r \\ 0 & \sin \theta & \cos \theta & -y_r \sin \theta - z_r \cos \theta + y_r \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta & z_r \sin \theta + y_r (1 - \cos \theta) \\ 0 & \sin \theta & \cos \theta & z_r (1 - \cos \theta) - y_r \sin \theta \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Practice Problem 3:

A cube is defined by 8 vertices:

A (0, 0, 0)

B (2, 0, 0)

C (2, 2, 0)

D (0, 2, 0)

E (0, 0, 2)

F (0, 2, 2)

G (2, 0, 2)

Perform the following 3D transformations on the cube:

i) Translation by $T = [t_x, t_y, t_z] = [5, 3, 4]$

ii) Scaling by $S = [s_x, s_y, s_z] = [1, 2, 0.5]$

iii) Rotation about X-axis by 90° in clockwise direction

Solution for Practice Problem 3:



Solⁿ : Let us compute the coordinates for asked operations.

1. Translation by $T = [5, 3, 4]$

Coordinates of translated cube are computed as,

$$P' = T \cdot P = \begin{bmatrix} 1 & 0 & 0 & t_x \\ 0 & 1 & 0 & t_y \\ 0 & 0 & 1 & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 2 & 2 & 0 & 0 & 0 & 2 & 2 \\ 0 & 0 & 2 & 2 & 0 & 2 & 0 & 2 \\ 0 & 0 & 0 & 0 & 2 & 2 & 2 & 2 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 & 5 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 2 & 2 & 0 & 0 & 0 & 2 & 2 \\ 0 & 0 & 2 & 2 & 0 & 2 & 0 & 2 \\ 0 & 0 & 0 & 0 & 2 & 2 & 2 & 2 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

Solution for Practice Problem 3:



$$P = \begin{bmatrix} 5 & 7 & 7 & 5 & 5 & 5 & 7 & 7 \\ 3 & 3 & 5 & 5 & 3 & 5 & 3 & 5 \\ 4 & 4 & 4 & 4 & 6 & 6 & 6 & 6 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

2. Scaling by $S = [1, 2, 0.5]$
Coordinates of Scaled Cube are computed as,
 $P' = S \cdot P$

$$= \begin{bmatrix} S_x & 0 & 0 & 0 \\ 0 & S_y & 0 & 0 \\ 0 & 0 & S_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 2 & 2 & 0 & 0 & 0 & 2 & 2 \\ 0 & 0 & 2 & 2 & 0 & 2 & 0 & 2 \\ 0 & 0 & 0 & 0 & 2 & 2 & 2 & 2 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

Solution for Practice Problem 3:



$$P = \begin{bmatrix} 0 & 2 & 2 & 0 & 0 & 0 & 2 & 2 \\ 0 & 0 & 4 & 4 & 0 & 4 & 0 & 4 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

3. Rotation about X-axis by 90° in clockwise direction

Here, rotation is in clockwise direction, so $\theta = -90^\circ$

Coordinates of scaled cube are computed as,

$$P' = R_{x(\theta=-90^\circ)} \cdot P$$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta & 0 \\ 0 & \sin\theta & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Solution for Practice Problem 3:



$$\begin{bmatrix} 0 & 2 & 2 & 0 & 0 & 0 & 2 & 2 \\ 0 & 0 & 2 & 2 & 0 & 2 & 0 & 2 \\ 0 & 0 & 0 & 0 & 2 & 2 & 2 & 2 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos(-90^\circ) & -\sin(-90^\circ) & 0 \\ 0 & \sin(-90^\circ) & \cos(-90^\circ) & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 2 & 2 & 0 & 0 & 0 & 2 & 2 \\ 0 & 0 & 2 & 2 & 0 & 2 & 0 & 2 \\ 0 & 0 & 0 & 0 & 2 & 2 & 2 & 2 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

Solution for Practice Problem 3:



$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 2 & 2 & 0 & 0 & 0 & 2 & 2 \\ 0 & 0 & 2 & 2 & 0 & 2 & 0 & 2 \\ 0 & 0 & 0 & 0 & 2 & 2 & 2 & 2 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

$$P_{ic} = \begin{bmatrix} 0 & 2 & 2 & 0 & 0 & 0 & 2 & -2 \\ 0 & 0 & 0 & 0 & 2 & 2 & 2 & 2 \\ 0 & 0 & -2 & -2 & 0 & -2 & 0 & -2 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

Practice Problem 4:



Find out the 3D transformation matrix to rotate a given 3D object by 60° about a line passing from point $(1, 1, 1)$ and the direction vector $V = 2i + 2j + 2k$

Solution for Practice Problem 4:



Here $\theta = 60^\circ$
Line is passing from $(1, 1, 1)$ and its direction vector is $V = 2i + 2j + 2k$. So line interpolates all points having the same x, y & z components hence line also passes through the origin.
 $V = 2i + 2j + 2k = i + j + k = ni + nj + nk$
So $(a, b, c) = (1, 1, 1)$
As derived in the previous section,
 $d = \sqrt{b^2 + c^2} = \sqrt{1+1} = \sqrt{2}$
And length of the vector,
 $l = |V| = \sqrt{a^2 + b^2 + c^2} = \sqrt{1+1+1} = \sqrt{3}$

Solution for Practice Problem 4:



Transformation matrix to rotate an object about the arbitrary line is given as,

$$M = R_x(\alpha) \cdot R_y(\beta) \cdot R_z(\theta) \cdot R_y(\beta) \cdot R_x(\alpha) \\ = A_v^{-1} \cdot R_z(\theta) \cdot A_v$$

Length of vector V is not normalized, so

$$R_x(\beta) \cdot R_x(\alpha) = A_v = \begin{bmatrix} d/|v| & -ab/(d \cdot |v|) & -ac/(d \cdot |v|) & 0 \\ 0 & c/d & -b/d & 0 \\ a/|v| & b/|v| & c/|v| & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \\ = \begin{bmatrix} \sqrt{2}/\sqrt{3} & -1/\sqrt{2}\sqrt{3} & -1/\sqrt{2}\sqrt{3} & 0 \\ 0 & 1/\sqrt{2} & -1/\sqrt{2} & 0 \\ 1/\sqrt{3} & 1/\sqrt{3} & 1/\sqrt{3} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Solution for Practice Problem 4:



$$R_z^{-1}(\alpha) \cdot R_y^{-1}(\beta) = A_V^T = \begin{bmatrix} \sqrt{2}/\sqrt{3} & 0 & 1/\sqrt{3} & 0 \\ 1/\sqrt{2}\sqrt{3} & 1/\sqrt{2} & 1/\sqrt{3} & 0 \\ -1/\sqrt{2}\sqrt{3} & -1/\sqrt{2} & 1/\sqrt{3} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$R_z(\theta = 60^\circ) = \begin{bmatrix} \cos \theta & -\sin \theta & 0 & 0 \\ \sin \theta & \cos \theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} \sqrt{3} & -\sqrt{3} & 0 & 0 \\ \sqrt{3} & \sqrt{3} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Solution for Practice Problem 4:



$$\begin{aligned} M &= A_v^T \cdot R_{z(\theta=60^\circ)} \cdot A_v \\ &= \begin{bmatrix} \sqrt{2}/\sqrt{3} & 0 & 1/\sqrt{3} & 0 \\ 1/\sqrt{2}\sqrt{3} & 1/\sqrt{2} & 1/\sqrt{3} & 0 \\ -1/\sqrt{2}\sqrt{3} & -1/\sqrt{2} & 1/\sqrt{3} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \sqrt{3} & -\sqrt{3} & 0 & 0 \\ \sqrt{3} & \sqrt{3} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} \sqrt{2}/\sqrt{3} & -1/\sqrt{2}\sqrt{3} & -1/\sqrt{2}\sqrt{3} & 0 \\ 0 & 1/\sqrt{2} & -1/\sqrt{2} & 0 \\ 1/\sqrt{3} & 1/\sqrt{3} & 1/\sqrt{3} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \end{aligned}$$

Multiplying last two matrices:

$$= \begin{bmatrix} \sqrt{2}/\sqrt{3} & 0 & 1/\sqrt{3} & 0 \\ 1/\sqrt{3}\sqrt{3} & 1/\sqrt{2} & 1/\sqrt{3} & 0 \\ -1/\sqrt{2}\sqrt{3} & -1/\sqrt{2} & 1/\sqrt{3} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Solution for Practice Problem 4:



$$\begin{bmatrix} \sqrt{2} & -1 - \frac{\sqrt{3}}{\sqrt{2}} & \frac{\sqrt{3} - 1}{\sqrt{2}} & 0 \\ \sqrt{2} & \frac{\sqrt{3} - 1}{\sqrt{2}} & \frac{\sqrt{3} - 1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$M = \begin{bmatrix} \frac{2\sqrt{3} + 1}{3} & -\frac{(\sqrt{3} - 2)}{3} & \frac{4 - \sqrt{3}}{3} & 0 \\ \frac{4 - \sqrt{3}}{3} & \frac{\sqrt{3} + 3}{6} & \frac{7\sqrt{3} - 7}{6} & 0 \\ -\frac{(\sqrt{3} - 2)}{3} & \frac{11 - 5\sqrt{3}}{6} & -\frac{(4 - 2\sqrt{3})}{6} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Practice Problem 5:



Rotate a triangle with vertices A (0, 2, 1), B (2, 3, 0) and C (1, 2, 1) by 45° around a line passing through the points (0, 0, 0) and (1, 1, 1).

Solution for Practice Problem 5:



olⁿ:

As derived in previous section,

$$D = \sqrt{b^2 + c^2} = \sqrt{(0-1)^2 + (0-1)^2} \sqrt{1+1} = \sqrt{2}$$

And length of the rotation axis,

$$\begin{aligned} l = |v| &= \sqrt{a^2 + b^2 + c^2} \\ &= \sqrt{(0-1)^2 + (0-1)^2 + (0-1)^2} \\ &= \sqrt{1+1+1} \\ &= \sqrt{3} \end{aligned}$$

The rotation axis is already passing through the origin, so there is no need for translating it. So the desired transformation sequence can be achieved by,

$$\begin{aligned} M &= R_x^{-1}(\alpha) \cdot R_y^{-1}(\beta) \cdot R_z(\theta) \cdot R_y(\beta) \cdot R_x(\alpha) \\ &= A v^{-1} \cdot R_z(\theta) \cdot A v \end{aligned}$$

Solution for Practice Problem 5:



where, $A_v = \begin{bmatrix} d/|v| & -ab/d \cdot |v| & -ac/d \cdot |v| & 0 \\ 0 & c/d & -b/d & 0 \\ a/|v| & b/|v| & c/|v| & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

$$= \begin{bmatrix} \sqrt{2}/\sqrt{3} & -1/\sqrt{6} & -1/\sqrt{6} & 0 \\ 0 & 1/\sqrt{2} & -1/\sqrt{2} & 0 \\ -1/\sqrt{3} & 1/\sqrt{3} & 1/\sqrt{3} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

 $R_z(\theta=45^\circ) = \begin{bmatrix} \cos \theta & -\sin \theta & 0 & 0 \\ \sin \theta & \cos \theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

Solution for Practice Problem 5:



$$= \begin{bmatrix} \cos 45^\circ & -\sin 45^\circ & 0 & 0 \\ \sin 45^\circ & \cos 45^\circ & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} & 0 & 0 \\ 1/\sqrt{2} & 1/\sqrt{2} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$M = A_v^{-1} \cdot R_z(0) \cdot A_v = \begin{bmatrix} \sqrt{2}/\sqrt{3} & 0 & 1/\sqrt{3} & 0 \\ -1/\sqrt{6} & 1/\sqrt{2} & 1/\sqrt{3} & 0 \\ -1/\sqrt{6} & -1/\sqrt{2} & 1/\sqrt{3} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} & 0 & 0 \\ 1/\sqrt{2} & 1/\sqrt{2} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Solution for Practice Problem 5:



$$\begin{aligned}
 & \begin{bmatrix} 0 & \sqrt{2}/\sqrt{3} & -1/\sqrt{6} & -1/\sqrt{6} & 0 \\ 0 & 0 & 1/\sqrt{2} & -1/\sqrt{2} & 0 \\ 0 & 1/\sqrt{3} & 1/\sqrt{3} & 1/\sqrt{3} & 0 \\ 1 & 0 & 0 & 0 & 1 \end{bmatrix} \\
 & M = \begin{bmatrix} 0.815 & 0 & 0.578 & 0 \\ -0.408 & 0.709 & 0.578 & 0 \\ -0.408 & -0.709 & 0.578 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \\
 & \begin{bmatrix} 0.709 & -0.709 & 0 & 0 \\ 0.709 & 0.709 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0.815 & -0.408 \\ 0 & 0 \\ 0.578 & 0 \\ 0 & 0 \end{bmatrix}
 \end{aligned}$$

Solution for Practice Problem 5:



$$\begin{bmatrix} 0.815 & -0.408 & 0.408 & 0 \\ 0 & 0.709 & 0.709 & 0 \\ 0.578 & 0.578 & 0.578 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Multiplying last two matrices,

$$M = \begin{bmatrix} 0.815 & 0.578 & 0 & 0 \\ -0.408 & 0.709 & 0.578 & 0 \\ -0.408 & -0.709 & 0.578 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0.578 & -0.792 & 0.213 & 0 \\ 0.578 & 0.213 & -0.792 & 0 \\ 0.578 & 0.578 & 0.578 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Solution for Practice Problem 5:



$$M = \begin{bmatrix} 0.805 & -0.311 & 0.508 & 0 \\ 0.508 & 0.808 & -0.0314 & 0 \\ -0.312 & 0.506 & 0.809 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Transformed coordinates,

$$P' = M \cdot P = \begin{bmatrix} 0.805 & -0.311 & 0.508 & 0 \\ 0.508 & 0.808 & -0.0314 & 0 \\ -0.312 & 0.506 & 0.809 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 2 & 1 \\ 2 & 3 & 2 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

Solution for Practice Problem 5:



$$= \begin{bmatrix} -0.114 & 0.677 & 0.691 \\ 1.302 & 3.440 & 1.810 \\ 1.821 & 0.894 & 1.509 \\ 1 & 1 & 1 \end{bmatrix}$$

Original Coordinates	Transformed Coordinates
A (0, 2, 1)	A' (-0.114, 1.302, 1.821)
B (2, 3, 0)	B' (0.677, 3.440, 0.894)
C (1, 2, 1)	C' (0.691, 1.810, 1.509)



COMPUTER AIDED ANALYSIS AND DESIGN

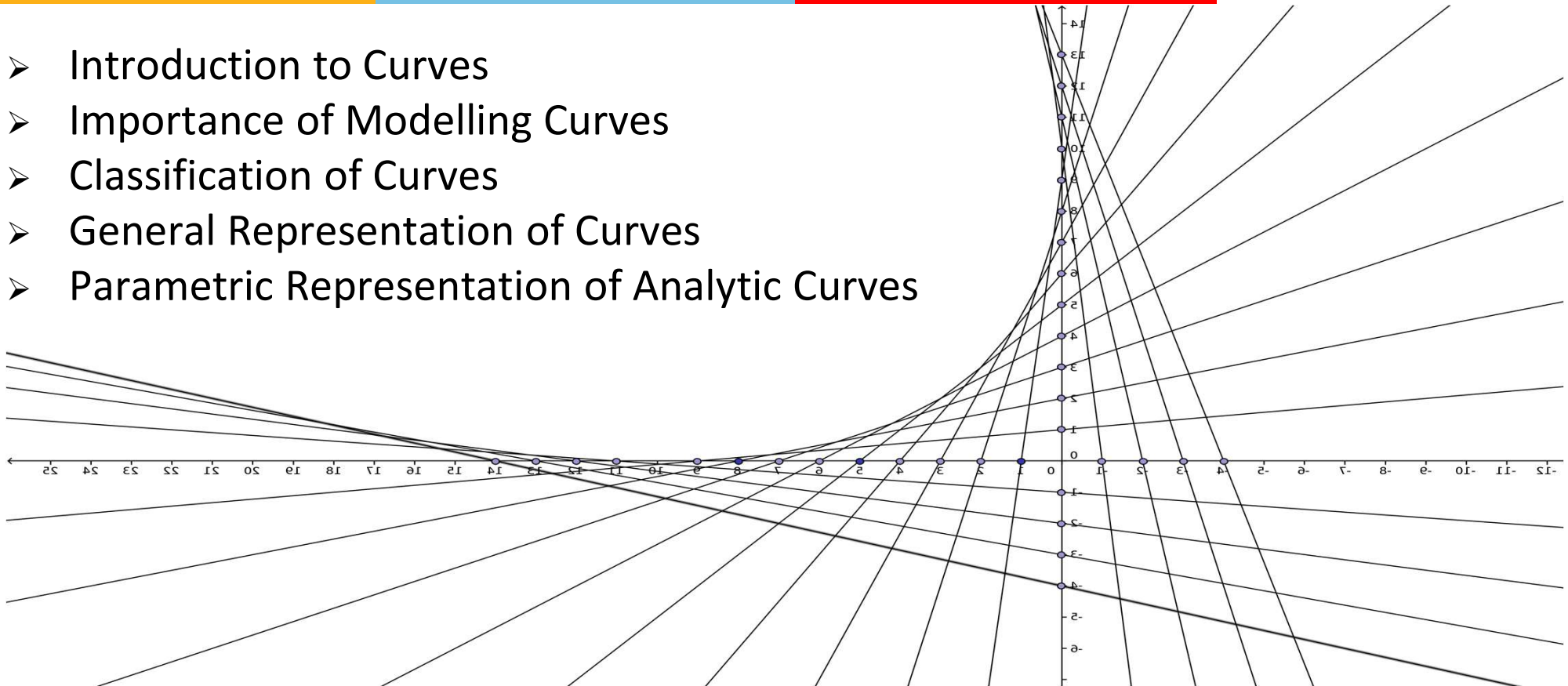
Course (DE ZG622) - Lecture 5

Prof. Ruchi More Kotecha

Session Outline

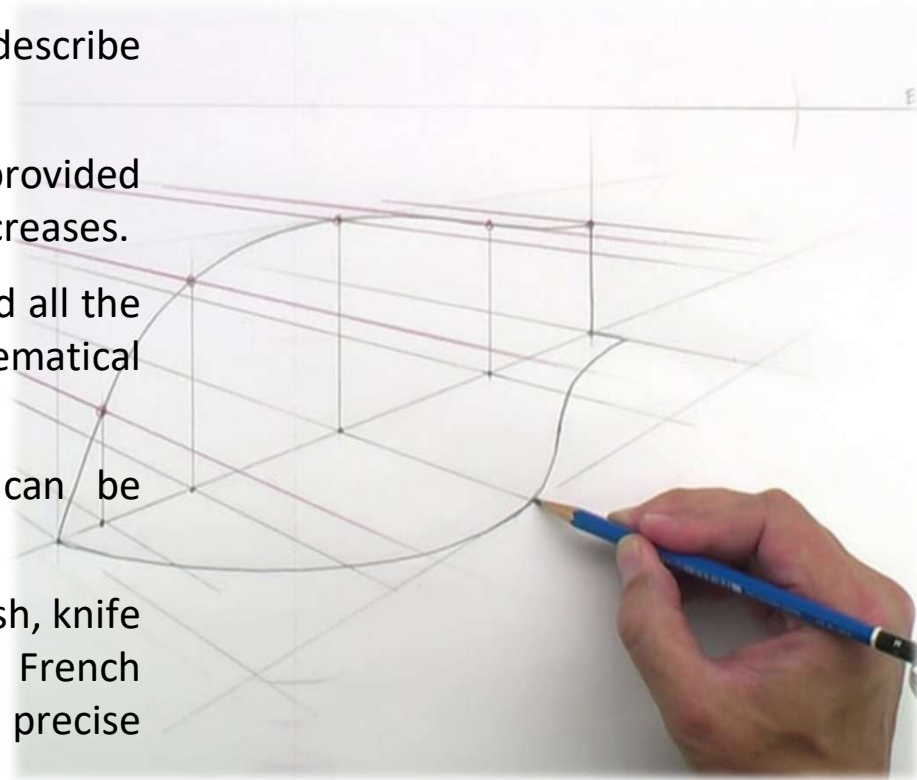


- Introduction to Curves
- Importance of Modelling Curves
- Classification of Curves
- General Representation of Curves
- Parametric Representation of Analytic Curves



What is a Curve?

- In mathematics, a curve is an abstract term used to describe the path of a continuously moving point.
- A curve may be presented by a collection of points, provided they are closely spaced as the radius of curvature decreases.
- Therefore, a curve is a collection of points (locus) and all the points on the curve satisfy certain mathematical representation (equation or behavior).
- Curves are basic entities from which surfaces can be constructed and in turn the shape of bodies.
- Curve can be created manually using pen/pencil, brush, knife or by using tools such as compass, straight scale, French curves, curve templates etc. However, this is not a precise representation.



Requirement for Modelling Curves



- In mathematics, a curve is an abstract term used to describe the path of a continuously moving point.
- Many real engineering designs need curved mechanical parts, civil engineering designs, architectural designs, aeronautics, ship building
- For modeling curves/surfaces in computer graphics they must be able to get represented mathematically so that their shapes can be controlled mathematically
- **The ability to control the shape is the most important requirement for a curve in CAD.**



Need for Mathematical Representation of Curves

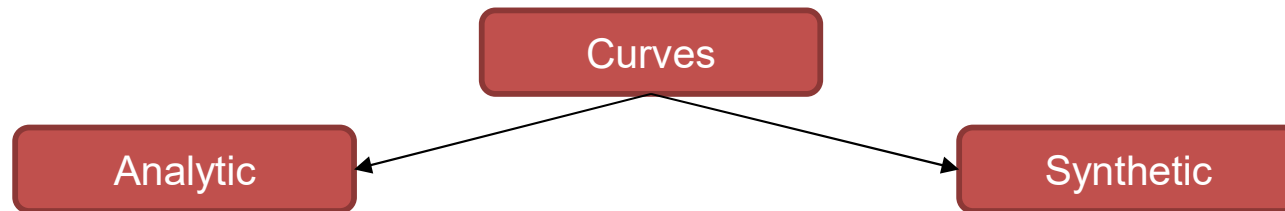
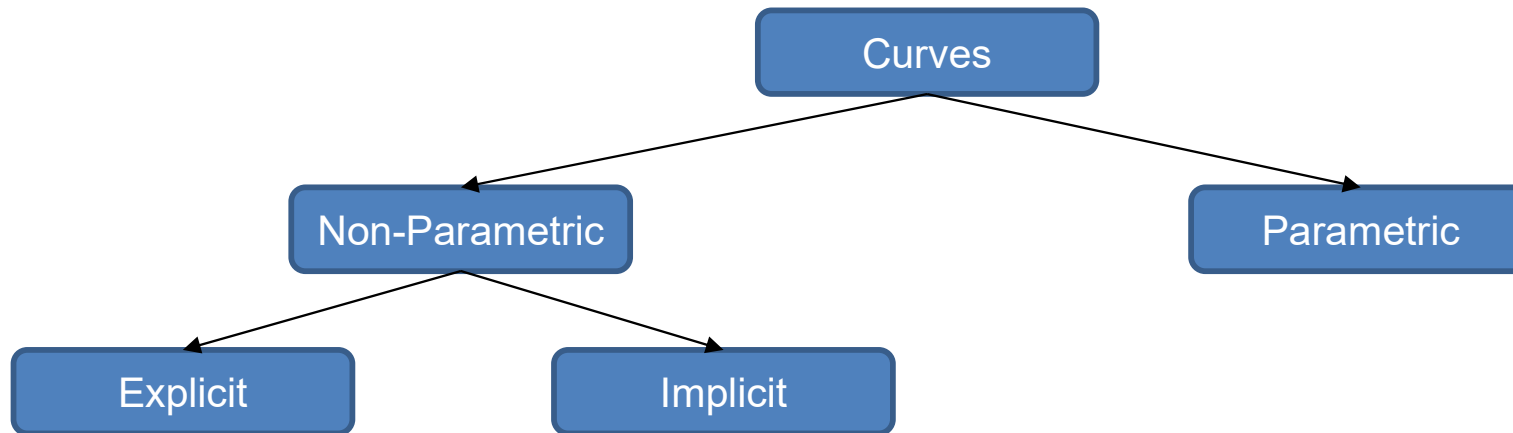


Advantages of mathematical representation of curves over its representation by a grid of points:

- A mathematical representation is precise, and the properties of the curve such as slope and radius of curvature can be easily calculated.
- A mathematical representation can be compactly stored in a computer.
- Drawings of an object represented mathematically in a computer can be easily produced.
- When curves are analytically defined over a region of interest, no interpolation scheme is needed to compute intermediate points.
- The use of points to represent curves has proven difficult when it is desirable to continually alter the shape of the curves to meet certain design criteria.
- It is very easy to compute any point on the curve by using its mathematical representation
- It is cumbersome to define curve by set points as it does not provide ability to control the shape



Classification of Curves





Non-Parametric Curves

Explicit Curves:

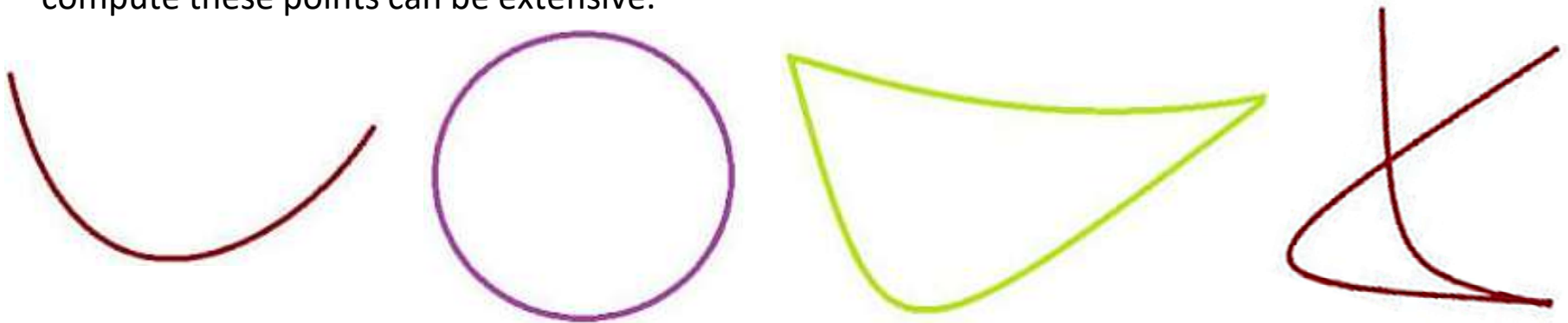
- An explicit curve is a graph of a function that's described by an equation that explicitly states the functional form.
- $y = f(x) + c$
- Here, dependent variable is expressed in terms of independent variable.
- Explicit curves cannot represent closed curves due to one-to-one relation between the dependent and independent variables.
- Explicit curves cannot represent vertical lines as they are single valued and due to the infinite nature of the slope.

Implicit Curves:

- An implicit curve is a function that can't be written as one variable in terms of the other variable.
- $f(x, y) + c = 0$
- No distinction can be made between dependent and independent variable.
- Implicit Curves can represent closed curves.
- Implicit curves fail the vertical line test (meaning that some values of x are associated with more than one value of y) and so are not necessarily graphs of functions.

Limitations of Non-Parametric Curves

- If the slope at certain point is vertical or almost vertical it can result into infinity condition which is difficult to handle programmatically. There can be other ill defined mathematical conditions that can arise.
- Shape of such curves are very much dependent on the coordinate system (axis dependent). In true sense, shape of the curve should be defined by relationship between data points and not by data points and some arbitrary coordinate system.
- If the curve display is achieved by series of points or line segments, then computation required to compute these points can be extensive.





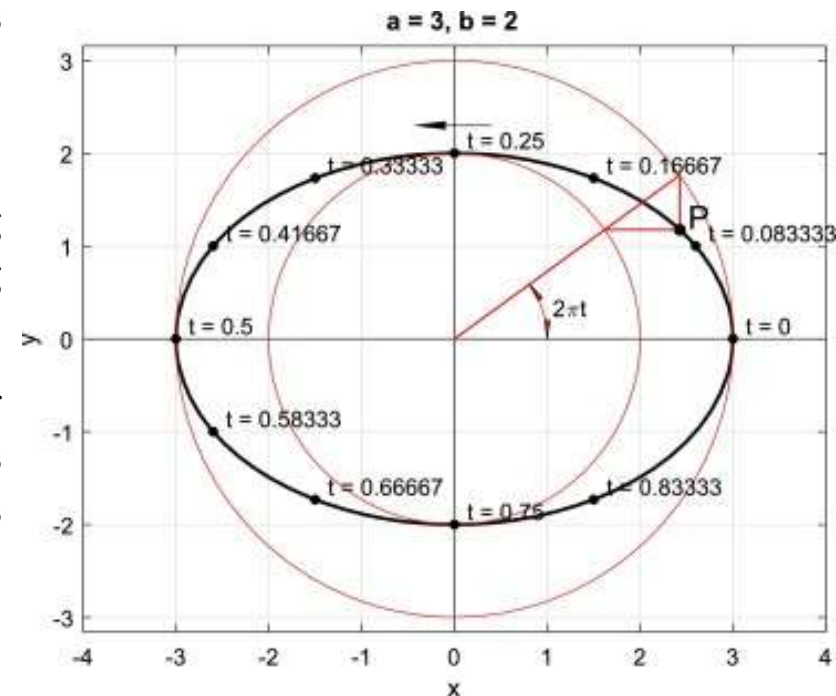
Parametric Curves

- A parametric curve has its defining equations in terms of simple, common, independent variable called the parametric variable.
- Each point on the curve is expressed as a function of this parameter. It is like a local coordinate for points on the curve.
- A curve segment is a bounded collection of points whose coordinates are given by continuous, one-parameter, single-valued functions like
- $x = x(u), y = y(u), z = z(u)$
- $u_{\min} \leq u \leq u_{\max}$
- Typically $u_{\min}$ and $u_{\max}$ are normalized to 0 and 1
- Positive sense of the curve is in the direction of increasing value of u .



Benefits of Parametric Curves

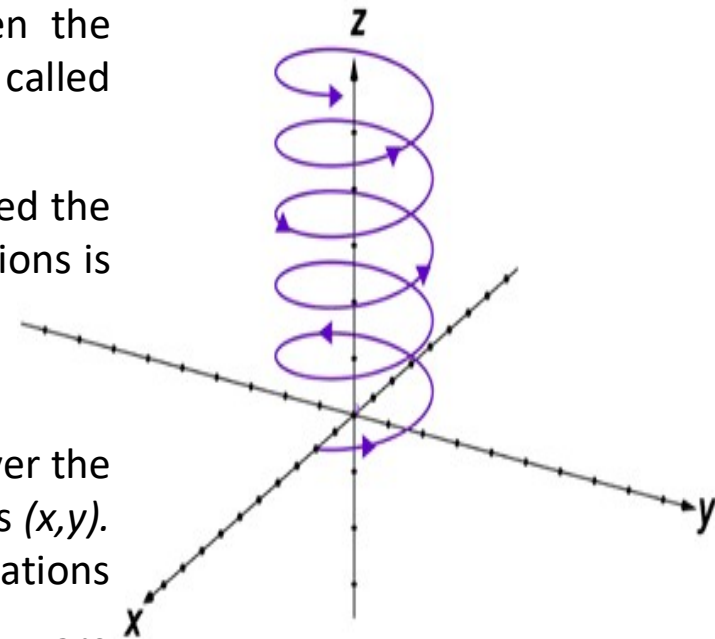
- It overcomes the limitations of non-parametric curves. It is suitable for representing closed and multivalued curves.
- Parametric curves can handle infinite slope condition.
- Solving for x , y and z coordinates simply involves solving polynomials in terms of parameter u rather than finding roots of equations $f(x,y) = 0$ which is more complex.
- Parametric form is more general and hence suited for computations and display and as the parametric form makes it easy to specify a start and end point, and generate points in between.
- If the parameter chosen represents time you'll have an extremely clearer intuition of where is the particle in a particular moment.
- Parametric curves are axis independent.





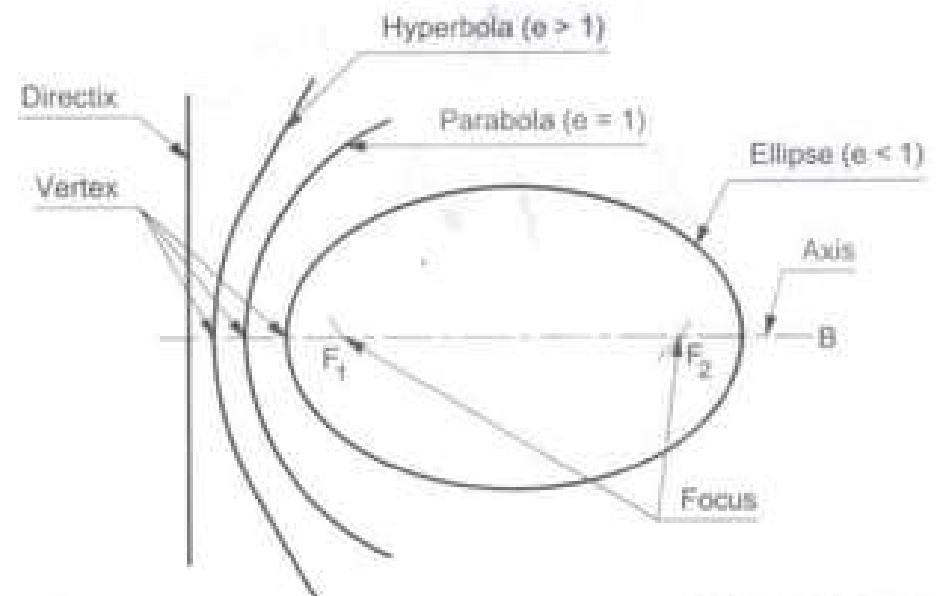
Representation of Parametric Curves

- If x and y are continuous functions of t on an interval I , then the equations $x=x(t)$ and $y=y(t)$ are called parametric equations and t is called the parameter.
- The set of points (x,y) obtained as t varies over the interval I is called the graph of the parametric equations. The graph of parametric equations is called a parametric curve or plane curve.
- Notice in this definition that x and y are used in two ways:
- The first is as functions of the independent variable t . As t varies over the interval I , the functions $x(t)$ and $y(t)$ generate a set of ordered pairs (x,y) . This set of ordered pairs generates the graph of the parametric equations
- In this second usage, to designate the ordered pairs, x and y are variables. It is important to distinguish the variables x and y from the functions $x(t)$ and $y(t)$.
- Watch: <https://www.youtube.com/watch?v=345SnWfahhY>



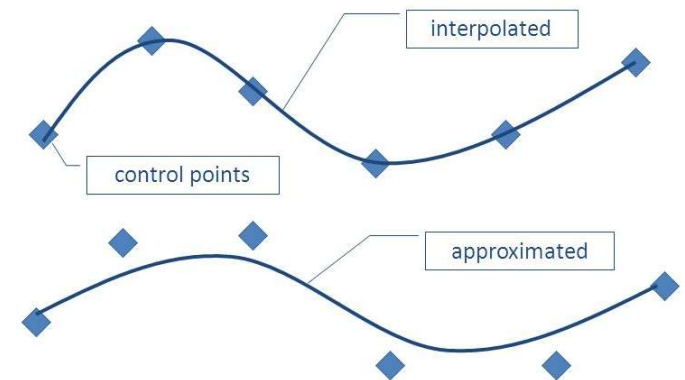
Analytic Curves

- Analytic curves are the ones that can be described by analytic equations such as that of lines, circles, conic sections etc.
- Analytic curves provide a very compact form to represent the shape.
- Computation of properties such as area, volume is relatively easy.
- Analytic curves cannot model complex shapes such as propeller blades, aircraft fuselage, ship hull etc.



Synthetic Curves

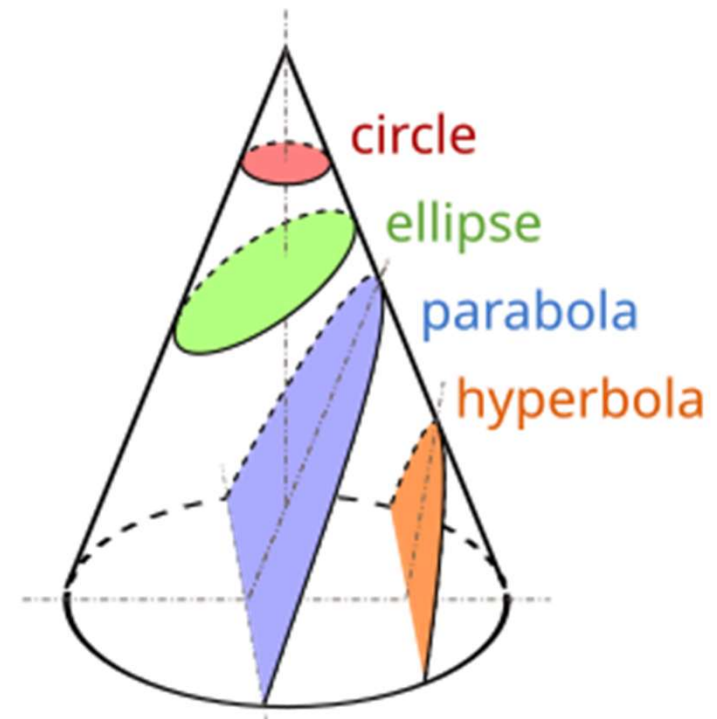
- Synthetic curves are the ones that are described by a set of data points called as control points.
- They offer flexibility in shape modeling and smoothness
- Eg. Splines, Bezier and B-spline curves
- These type of curves are used to model complex curves and surfaces required in real life applications such as propeller blades, ship hull, car body design, aircraft fuselage design etc.



Parametric Representation of Analytic Curves



- Line
- Circle
- Ellipse
- Cylinder
- Helix
- Archimedean Spiral

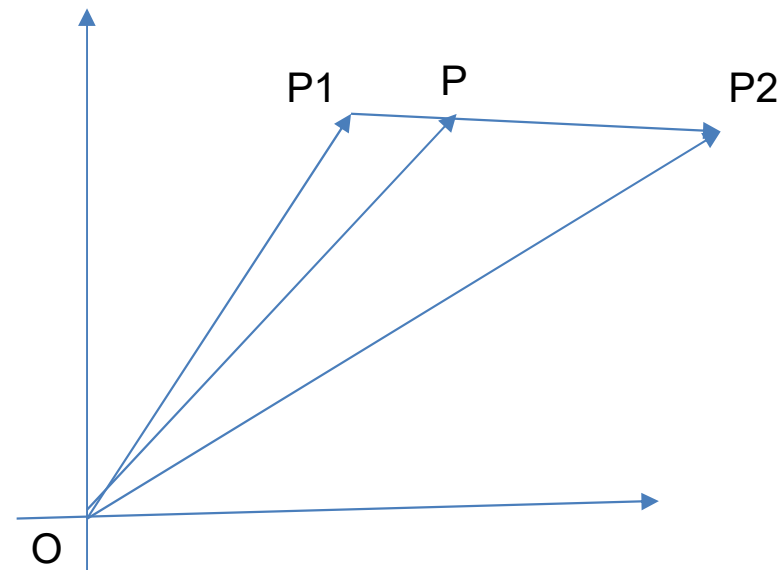


Recall of Conic Sections

Parametric Equation of a Line

- $P = P_1 + (P - P_1)$
- Let $u = (P - P_1)/(P_2 - P_1)$
- So $(P - P_1) = u (P_2 - P_1)$
- Hence parametric equation of line is

$$P = P_1 + u(P_2 - P_1) \quad \text{where } 0 \leq u \leq 1$$
- If $u = 0$, P coincides with P_1
- If $u = 1$, P coincides with P_2
- So scalar form of equations of line are:
- $x = x_1 + u(x_2 - x_1)$
- $y = y_1 + u(y_2 - y_1)$
- in 3D, add : $z = z_1 + u(z_2 - z_1)$

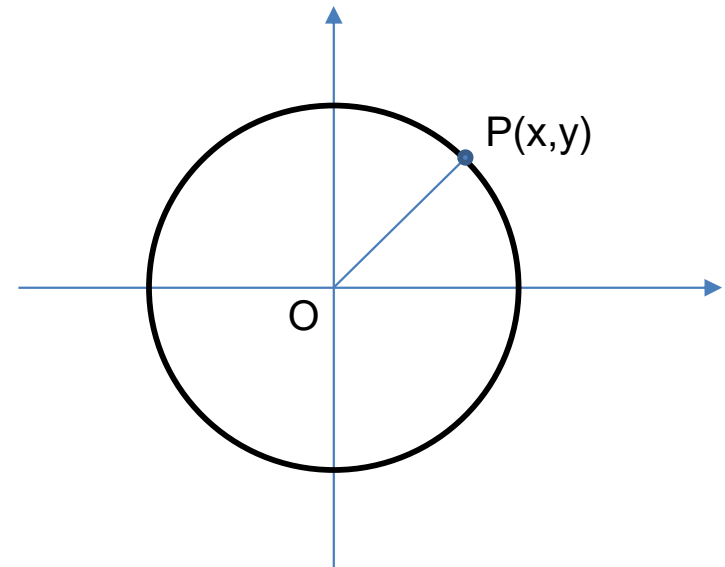




Parametric Equation of a Circle

When Center is at the Origin

- $x^2 + y^2 = r^2$
- Parametric form:
 - $x = r \cos \theta$
 - $y = r \sin \theta$where $0 \leq \theta \leq 2\pi$
- Or
 - $x = r \cos 2\pi u$
 - $y = r \sin 2\pi u$where $0 \leq u \leq 1$





Parametric Equation of a Circle

When Center is at (x_c, y_c)

➤ $(x - x_c)^2 + (y - y_c)^2 = r^2$

➤ Parametric form:

➤ $x = x_c + r\cos\theta$

where $0 \leq \theta \leq 2\pi$

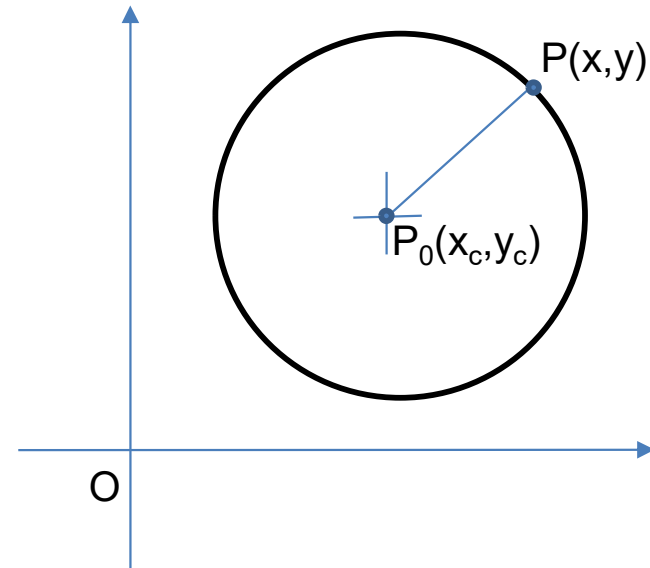
➤ $y = y_c + r\sin\theta$

Or

➤ $x = x_c + r\cos 2\pi u$

where $0 \leq u \leq 1$

➤ $y = y_c + r\sin 2\pi u$



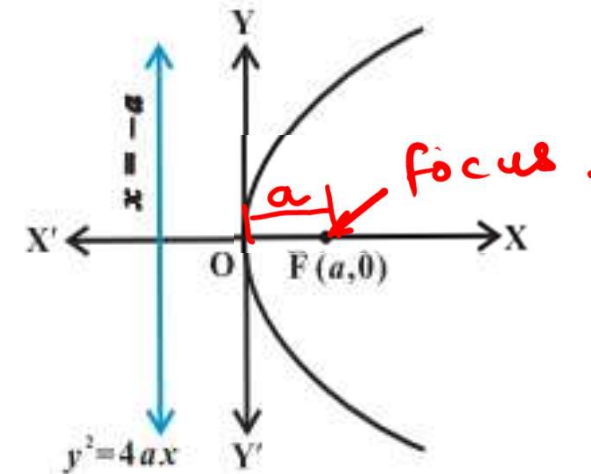
Parametric Equation of a Parabola

- Explicit form:
 $y^2 = 4ax$ (with vertex at origin)

- Parametric form:

$$x = a\theta^2$$

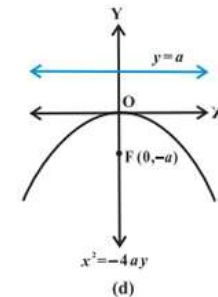
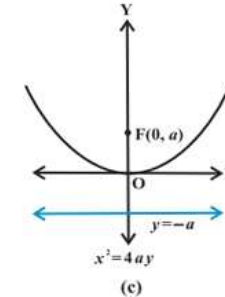
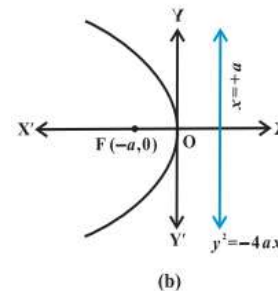
$$y = 2a\theta$$



where,

a = Focal Length

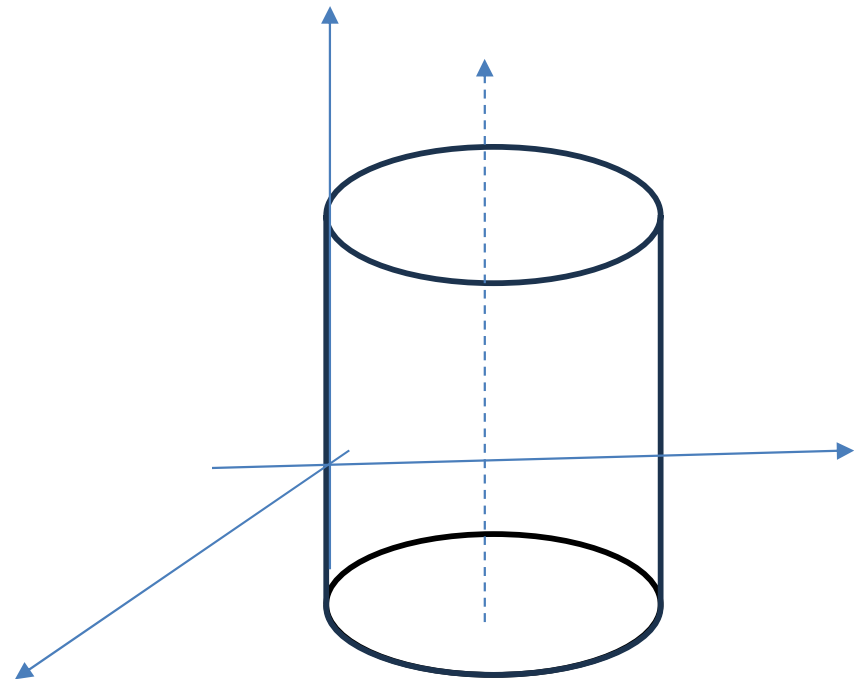
Focus is at $(a, 0)$



Parametric Equation of a Cylinder

When Center is at (x_c, y_c)

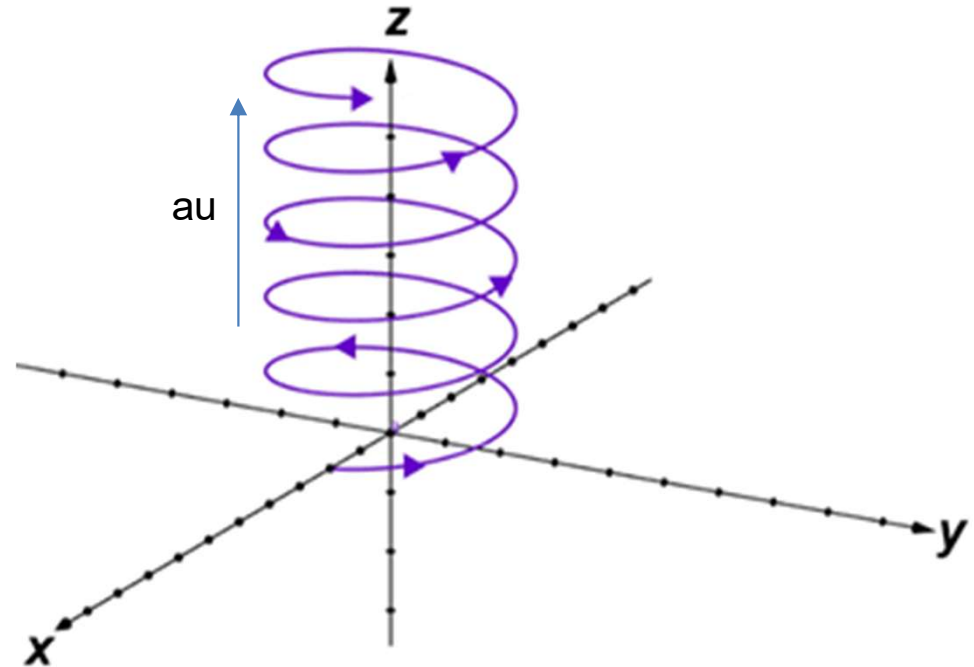
- Parametric form:
- $x = x_c + r \cos 2\pi u$ where $0 \leq u \leq 1$
- $y = y_c + r \sin 2\pi u$
- $z = hv$ where $0 \leq v \leq 1$



Parametric Equation of a Helix

When Center is at Origin

- Parametric form:
- $x = r \cos n2\pi u$ where $0 \leq u \leq 1$
- $y = r \sin n2\pi u$
- $z = au$ where a is any constant



Parametric Equation of an Archimedean Spiral



- Parametric form:
 - $x = (a+b\theta)\cos\theta$
 - $y = (a+b\theta)\sin\theta$
- where $0 \leq \theta \leq 2\pi$

Or

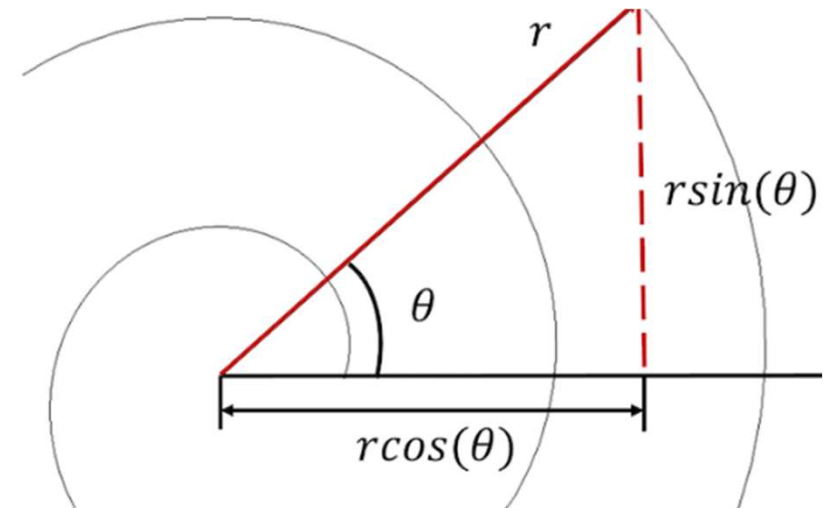
- $x = (a+b2\pi u)\cos2\pi u$
 - $y = (a+b2\pi u)\sin2\pi u$
- where $0 \leq u \leq 1$

Where,

a = initial radius

b = pitch distance i.e. $(\text{final radius} - \text{initial radius})/(2\pi n)$

n = Number of convolutions



Recursive relations for finding points on the curves - Circle



$$\cos(A+B) = \cos A \cos B - \sin A \sin B$$

$$\sin(A+B) = \sin A \cos B + \cos A \sin B$$

$$x_i = r \cos \theta \quad y_i = r \sin \theta$$

$$x_1 = r \cos \theta \quad y_1 = r \sin \theta$$

$$x_2 = r \cos(\theta + d\theta) \quad y_2 = r \sin(\theta + d\theta)$$

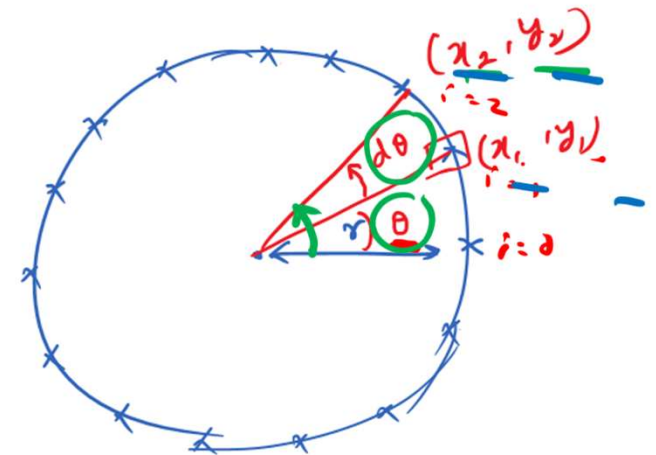
$$x_2 = r \cos \theta \cdot \cos d\theta - r \sin \theta \cdot \sin d\theta$$

$$x_2 = x_1 \cdot \cos d\theta - y_1 \cdot \sin d\theta$$

$$y_2 = r \sin \theta \cdot \cos d\theta + r \cos \theta \cdot \sin d\theta$$

$$y_2 = y_1 \cdot \cos d\theta + x_1 \cdot \sin d\theta$$

$$[x_{i+1}, y_{i+1}] = [x_i \cdot \cos d\theta - y_i \cdot \sin d\theta, y_i \cdot \cos d\theta + x_i \cdot \sin d\theta]$$



Recursive relations for finding points on the curves - Ellipse



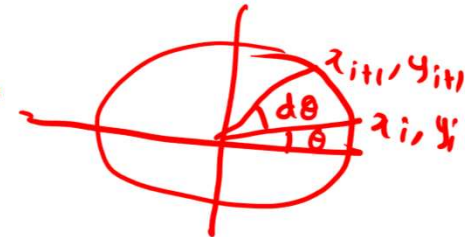
$$x_i = a \cos \theta$$

$$y_i = b \sin \theta$$

$$\begin{aligned} x_{i+1} &= a \cos(\theta + d\theta) \\ &= \underbrace{a \cos \theta}_{x_i} \cdot \cos d\theta - \underbrace{\frac{a}{b} \sin \theta}_{\frac{y_i}{b}} \cdot \sin d\theta \\ &= \underline{x_i} \cdot \cos d\theta - \frac{a}{b} \underline{y_i} \sin d\theta \end{aligned}$$

$$[x_{i+1}, y_{i+1}] :$$

$$\begin{aligned} y_{i+1} &= b \sin(\theta + d\theta) \\ &= \underbrace{b \sin \theta}_{y_i} \cdot \cos d\theta + \frac{b}{a} \underbrace{\cos \theta}_{x_i} \cdot \sin d\theta \\ &= \underline{y_i} \cdot \cos d\theta + \frac{b}{a} \underline{x_i} \sin d\theta \end{aligned}$$



Recursive relations for finding points on the curves - Parabola

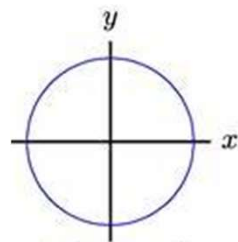


$$\begin{aligned} \underline{x_i} &= a\theta_i^2, \quad y_i = \pm (2a\theta_i) \\ x_{i+1} &= a(\theta_i + d\theta)^2 \\ &= \boxed{a\theta_i^2} + 2a\theta_i \cdot d\theta + a(d\theta)^2 \end{aligned}$$

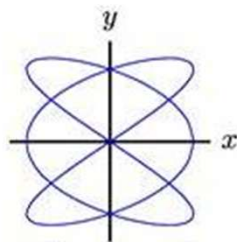
$$x_{i+1} = x_i + y_i \cdot d\theta + a \cdot (d\theta)^2$$

$$\begin{aligned} y_{i+1} &= \pm 2a(\theta_i + d\theta) \\ &= \boxed{\pm 2a\theta_i} + 2ad\theta \\ &= y_i + 2ad\theta \end{aligned}$$

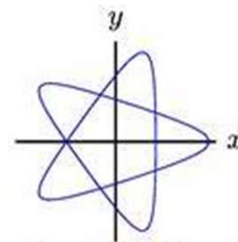
Shape Modeling Using Parametric Representation of Curves



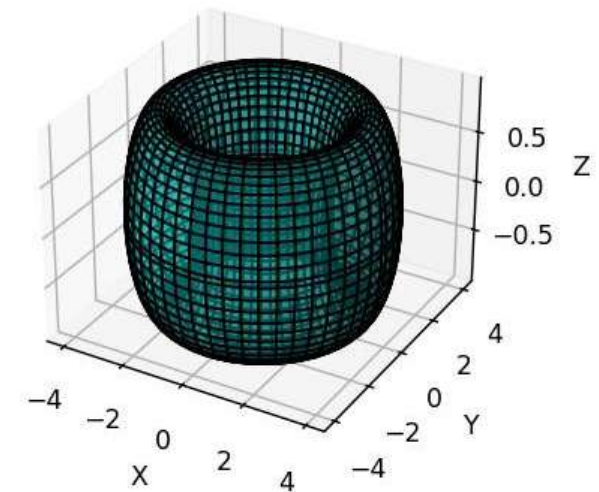
$$\begin{aligned}x(t) &= \cos(t) \\ y(t) &= \sin(t)\end{aligned}$$



$$\begin{aligned}x(t) &= 3 \cos(t) \\ y(t) &= 2 \sin(t)\end{aligned}$$



$$\begin{aligned}x(t) &= 2 \cos(x) + 5 \cos\left(\frac{2}{3}x\right) \\ y(t) &= 2 \sin(x) + 5 \sin\left(\frac{2}{3}x\right)\end{aligned}$$



[Cool Parametric Equations Animation | Manim \(youtube.com\)](https://www.youtube.com/watch?v=Mwe2Q86YfF0)

<https://www.youtube.com/watch?v=Mwe2Q86YfF0>



Thank You

Reference Question Bank for Mid Sem Exam. Questions asked in exam are not restricted to this question bank only. This question bank does not contain sums/ problems. Please refer class notes and textbooks for the same.

Introduction to CAD and Product life cycle

What are the phases through which a product typically goes over its life cycle?

State role of CAD, CAM and CAE in the life cycle of a product with diagram.

What are different coordinate systems used in geometric modeling? Explain with a diagram.

What is feature based geometric modeling? Explain with a diagram.

State and explain in detail the feature creation tools in solid modelling.

What are different geometric modeling approaches? Explain in detail.

Explain the functional areas of a basic CAD system and their applications in the design process.

State the role of computers in design analysis and manufacturing sectors.

2D and 3D Transformations

What are different transformations and their matrices? Categorize them as per their type.

What is the need of homogeneous coordinate system?

State general transformation matrix for 3D transformations in homogeneous coordinate system and explain the role of elements in it.

Explain the procedure to rotate 3D object about any arbitrary line.

State 2D transformation matrix for rotation about Z axis by an angle of 30 degrees in clockwise direction.

State resultant 2D transformation matrix when a position vector is elongated 2 times along X and Y direction and then mirrored about Y axis.

Geometric Modeling of Curves

What are the modeling requirements while using curves?

How curves are classified? Explain any two types of curves.

What is the difference between non-parametric and parametric curve?

What are explicit and implicit curves? Explain the advantages and limitations of each

What are limitations of non-parametric curves?

What benefits parametric curves offer?

Explain how successive points on the circle can be recursively found in terms of previous point from a given point $(x_1, y_1) = (r \cos \theta, r \sin \theta)$

Find out parametric equation of a helix with radius r and height h .

Cubic Spline or Hermite Curve

What are the characteristics of synthetic curves?

Explain C_0 , C_1 and C_2 continuity of curves. Draw a diagram showing C_2 continuity of 2 curves joining at a point.

What are the benefits of Cubic Spline curve?

What are limitations of Cubic Spline Curve?

How a Cubic Spline curve is mathematically represented in parametric form?

State the conditions by which the basis functions of a Cubic Spline can be determined.

Explain the parametric equation for a Cubic Spline curve in detail

Explain with the help of a diagram effect of change of direction and magnitude of tangent vectors on the shape of Hermite curve.

Explain chord length approximated parametric Cubic Spline curve. state the equations for finding the limits of the parameter.

What is normalized parametric Cubic Spline? State their properties.

What are different alternative end conditions that can be applied to Cubic Spline curve?

Bezier curve

State any 4 important properties of Bezier curve

How do Bezier curves overcome the limitations of Hermite curves?

What benefits a Bezier curve provides while modeling the shape?

What are the limitations of the Bezier curve?

Write the parametric equation of Bezier curve that has 5 control points.

Write basis functions of above Bezier curve.

How a point on the Bezier curve at certain value of parameter u can be obtained without evaluating its parametric equation?